Semiparametric Estimation of Optimal Reserve Prices with Generative Deep Learning

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November 8, 2026

Abstract

This paper studies inference for optimal reserve prices when only auction maxima are observed. Under symmetric independent private values, maxima identify the latent value distribution, which is estimated globally by a structural Wasserstein generative adversarial network with gradient penalty (WGAN-GP). I derive a cross-fitted Neyman-orthogonal score for a fixed-bandwidth regularized reserve. Primitive conditions cover training-fold empirical kernel moments and a fixed-pseudo-count calibration, with an explicit single-crossing target and failure rule. These estimators are first-order equivalent to a same-kernel full-sample plug-in. Inference can use either a Wald interval or the full set obtained by inverting the studentized score; a conservative interval envelope is reported separately and does not replace that set. Fresh validation with 100 repetitions at each of two sample sizes finds a modest small-sample coverage gain from score inversion, but does not establish a reliable advantage over Wald inference or an incremental gain from the GAN. Two small-sample replications produce disconnected, boundary-touching sets because sparse lower-tail information makes score variance highly sensitive to individual observations. All components and diagnostic flags are retained. Earlier experiments illustrate how good global WGAN fit can coexist with inaccurate local moments; they remain distinct from the fresh validation. Inference for the exact unsmoothed reserve requires additional bandwidth and bias conditions and is treated as an extension.

Keywords: optimal auctions; reserve prices; auction maxima; order statistics; double machine learning; cross-fitting; Neyman orthogonality; Wasserstein generative adversarial networks; semiparametric inference; Monte Carlo simulation.

JEL classification: C14; C15; C45; D44.

1 Introduction

Optimal-auction theory characterizes the seller's revenue-maximizing reserve price as a functional of the latent distribution of bidder valuations [Myerson, 1981]. In empirical settings, however, researchers and sellers often observe only a transaction price, a winning bid, or a small number of order statistics rather than the complete vector of private values. This paper asks whether a flexible generative model can recover the economically relevant features of the latent distribution while still permitting valid inference on the reserve-price parameter.

This question is difficult for a reason that is easy to hide with global fit statistics. If only the highest of N valuations is observed, then the probability of seeing an auction maximum below the reserve is $F_0(r_0)^N$. The reserve can therefore depend on a region with very few observations even when the observable maximum-bid distribution is well approximated overall. A small finite-sample fitting criterion is not a certificate of the local accuracy required by the optimal-reserve score. For a fixed bandwidth, a true population Wasserstein distance does bound the smoothed local moments through their Lipschitz constants; those constants deteriorate as the bandwidth shrinks.

The proposed framework treats the valuation distribution as an infinite-dimensional nuisance parameter and the reserve price as the low-dimensional parameter of interest. A structural WGAN-GP reproduces the auction observation process: it generates independent bidder values, applies the relevant order-statistic map, and matches the resulting observable distribution to the data. Cross-fitting separates nuisance estimation from score evaluation, while a Neyman-orthogonal correction reduces the first-order sensitivity of the reserve-price estimator to errors in the learned distribution.

The paper makes three contributions. First, it connects adversarial structural estimation to the order-statistics approach in empirical auctions. The generator respects the independent-private-values structure before applying the same maximum operator observed in the data. Second, it derives an observation-level orthogonal score for a smoothed reserve and gives conditions for cross-fitted asymptotic linearity and valid fixed-bandwidth inference. It distinguishes the full confidence set obtained by inverting the studentized score from a conservative interval envelope that contains it. For the empirical-local and fixed-pseudo-count calibrated learners, bounded kernel classes and a separated single-crossing root supply primitive conditions. The resulting estimators are first-order equivalent to the same-kernel full-sample plug-in; no first-order efficiency gain from the GAN is claimed. A pure WGAN learner requires additional local-rate conditions. A covariance-aware Richardson combination remains a conditional extension from that regularized target to the exact reserve. Third,

it uses nested Monte Carlo ablations to distinguish failures of global WGAN learning, local nuisance rates, smoothing bias, and empirical root selection rather than attributing all errors to the generative model.

The archived Monte Carlo results sharpen the role of WGAN-GP in this problem. It reproduces the observable maximum distribution reasonably well but misses local tail moments that matter for the reserve. Replacing only those score nuisances with training-fold empirical moments restores near-oracle coverage for the fixed-bandwidth target, and the paper verifies the required rate for that learner under a local overlap condition. For the exact reserve, a stable $n^{-0.2}$ bandwidth path combined with Richardson extrapolation reduces bias, and a three-bandwidth residual-bias envelope attains 96.5 percent coverage in the reported design. More aggressive shrinkage creates multiple roots; taking the union over all root chains increases interval length without improving coverage.

An archived paired follow-up implements a simple hybrid by giving the WGAN the weight of 50 pseudo-observations in the training-fold local moments. It substantially reduces error relative to the raw WGAN score without changing the network or loss, but tracks empirical-only estimation closely and does not establish an incremental WGAN advantage. A subsequent nested-validation ablation tests an eleven-parameter monotone generator and adaptive mixing separately and jointly. Its mixed results do not establish a uniform gain over empirical benchmarks. A retrospective initialization-only control nearly reproduces the lognormal results, so those gains cannot be attributed to adversarial training alone. The primary new validation fixes the single-crossing safeguards and score-inversion rule before drawing 100 fresh samples at each of $n = 500$ and $n = 2000$. At the smaller sample size, calibrated-GAN coverage changes from 91 percent under Wald inference to 93 percent under score inversion, with similarly modest changes for the empirical comparators. The small number of discordant pairs does not establish a reliable improvement. Two samples produce disconnected sets for the calibrated and empirical-local methods. All components are retained, and boundary and disconnection flags remain visible when a conservative interval envelope is supplied. At $n = 2000$, both interval constructions are close to the nominal level. The new results support a transparent reporting rule, not a claim of uniformly accurate finite-sample coverage or an incremental GAN advantage.

The formal coverage result concerns the fixed-bandwidth regularized reserve. Exact-reserve intervals remain a bias-aware prototype whose uniform validity requires additional conditions on the shrinking-bandwidth root process. The older simulation tables precede the stricter numerical gates and are retained separately from the independent validation rather

than pooled with it.

2 Related Literature

The starting point is the virtual-value characterization of optimal reserves in Myerson [1981]. The observation model belongs to the order-statistics literature surveyed by Athey and Haile [2007]: under symmetric independent private values and known competition, the distribution of the observed maximum identifies the latent valuation distribution. Menzel and Morganti [2013] develop large-sample theory for auction estimators based on order statistics. Relative to direct inversion, the present paper asks whether a structural generator can regularize the latent distribution while preserving inference on one economically important functional. The weakly observed lower tail of the maximum distribution is central rather than incidental to that comparison.

The computational component builds on Wasserstein generative adversarial networks [Arjovsky et al., 2017] and gradient-penalty training [Gulrajani et al., 2017]. It is also related to adversarial structural estimation, which embeds economic restrictions and a simulator inside a distribution- matching problem [Kaji et al., 2026]. Here the restriction is imposed before the observable is formed: independent bidder-level draws are mapped into auction maxima. The main diagnostic result is that good fit of this observable distribution does not guarantee the local CDF and density accuracy needed by the reserve score.

Finally, the inference layer follows the cross-fitting and orthogonality logic of double machine learning [Chernozhukov et al., 2018]. The target differs from standard treatment-effect and regression examples in two ways. It is an implicit nonlinear policy functional of a latent distribution, and its regularized score contains a density component whose variance changes with the bandwidth. The paper therefore separates three claims that are often combined: inference for a fixed regularized root, approximation of the exact optimal reserve as the bandwidth shrinks, and root-robust inference when the empirical moment is nonmonotone.

3 Auction Model and Target Parameter

Consider auctions indexed by $j = 1, \ldots, n$. In the baseline model each auction has the same known number $N \geq 2$ of risk-neutral bidders; N remains fixed as the number of auctions grows. Bidder valuations satisfy

$$V_{ij} \overset{\text{iid}}{\sim} F_0, \qquad i = 1, \ldots, N, \tag{1}$$

independently across auctions. Values are private, utilities are quasilinear, and the second-price allocation and payment rules are enforced. The seller's value from retaining the object is normalized to zero. The distribution F_0 is continuously differentiable on the positive real line with density f_0 and finite first moment. The observed data are $O_j = (B_j, N)$, where

$$B_j = \max_{1 \le i \le N} V_{ij}. \tag{2}$$

Under truthful bidding in the second-price benchmark, B_j is the highest submitted bid. It must be observed without censoring or truncation for every sampled auction, including auctions that do not result in a sale. It is not the transaction price, which is the second-highest bid when the reserve does not bind. The fixed bidder population and truthful-bidding conditions are maintained restrictions, not conclusions from a good distributional fit. Neither selective entry nor reserve-dependent participation is modeled here.

Assume a regular valuation distribution with a unique interior revenue-maximizing reserve $r_0 > 0$ and $f_0(r_0) > 0$. Its first-order condition is

$$1 - F_0(r_0) - r_0 f_0(r_0) = 0. \tag{3}$$

Equivalently, r_0 maximizes the single-bidder posted-price revenue $r\{1 - F_0(r)\}$. For N bidders, expected revenue in a second-price auction with common reserve r is

$$\mathcal{R}_N(r; F_0) = r\{1 - F_0(r)^N\} \tag{4}$$

$$+ \int_r^\infty \left[1 - F_0(t)^N - N\{1 - F_0(t)\}F_0(t)^{N-1}\right] dt. \tag{5}$$

The economic target is the scalar r_0. The main inferential result below instead concerns a precisely defined fixed-bandwidth regularized target, $\theta_{0,h}$. Point identification of r_0 does not by itself imply regular root-n inference for it. Connecting the two targets requires additional smoothness and a separate shrinking-bandwidth argument.

4 Identification from Order Statistics

Let $H_0(\cdot \mid N)$ denote the distribution of the observed maximum. Under the baseline assumptions,

$$H_0(b \mid N) = F_0(b)^N. \tag{6}$$

For known N, the latent distribution is therefore identified by

$$F_0(v) = H_0(v \mid N)^{1/N}. \tag{7}$$

Where $H_0(v \mid N) > 0$, writing $g_0 = \partial_v H_0$ also gives

$$f_0(v) = \frac{g_0(v \mid N)}{N H_0(v \mid N)^{1-1/N}}. \tag{8}$$

These are restrictions on the distribution of the complete highest value, not on a sale price or a maximum observed only after exceeding an existing reserve. In particular, conditioning the sample on sale changes the observed CDF and does not preserve the displayed inversion without further work.

Let $\widehat{H}_n$ be the empirical CDF of the observed maxima. The direct inversion estimator is

$$\widehat{F}_{\mathrm{DI}}(v) = \widehat{H}_n(v)^{1/N}. \tag{9}$$

It yields a density-free reserve estimator

$$\widehat{r}_{\mathrm{DI}} \in \arg \max_{r \in \{B_1, \ldots, B_n\}} r\{1 - \widehat{H}_n(r-)^{1/N}\}. \tag{10}$$

The candidate set is the observed positive prices, and $\widehat{H}_n(r-)$ counts maxima strictly below r. This left-limit convention treats a value equal to the reserve as eligible to trade and handles tied bids consistently. It is an explicit finite-candidate policy rather than an unattained supremum at a downward jump of a right-continuous empirical revenue curve. This direct estimator is the first benchmark because it separates identification from the approximation and optimization error introduced by WGAN-GP.

The transformation also makes the information problem transparent. When $F_0(r_0)$ is below one, the probability of observing a maximum below the reserve is $F_0(r_0)^N$, which can be small even for moderate N. Thus, tail and quantile errors near the economically relevant reserve can be substantial in finite samples despite point identification. The later semiparametric analysis asks whether the scalar reserve can nevertheless be estimated and inferred on through a regularized target without requiring uniformly accurate recovery of the entire distribution. It does not assert root-n inference for the unsmoothed reserve solely because that reserve is one-dimensional.

These conclusions do not automatically extend to unknown competition, correlated bidder values, auction-level unobserved heterogeneity, endogenous entry, or data containing the transaction price rather than the highest bid. The baseline also does not pool different bidder counts or heterogeneous auction populations under one common F_0 without an additional restriction.

5 WGAN-GP Nuisance Estimation

For the baseline implementation, the generator maps independent bidder-level noise into a log valuation:

$$\widetilde{L}_{ij} = G_\gamma(Z_{ij}), \qquad \widetilde{V}_{ij} = \exp(\widetilde{L}_{ij}), \qquad Z_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0, I). \tag{11}$$

The same scalar generator is called independently for every bidder. This is essential: a network that produces the entire bidder vector from one common noise draw generally induces dependence across values and violates the independent-private-values benchmark. Synthetic observables are constructed using the same order-statistic map as the data,

$$\log \widetilde{B}_j = \max_{1 \le i \le N} \widetilde{L}_{ij}. \tag{12}$$

A critic compares standardized real and synthetic log maxima. Gradient-penalty regularization imposes an approximate Lipschitz restriction on the critic. In the pilot, each network has two hidden layers with 64 units, the bidder-level noise dimension is eight, and every generator update follows five critic updates. Checkpoints are selected using only the empirical Wasserstein distance between real and generated maxima; the true reserve is not used in training or model selection.

The trained generator induces estimates of the latent distribution, revenue curve, and the smoothed maximum-bid moments required by the reserve-price score. In the DML pilot, nuisance training is repeated separately within each training fold and the score is evaluated only on the corresponding held-out fold.

Under the baseline without covariates, the score's two local moments are also directly estimable from observed maximum bids. The empirical-local ablation therefore estimates (A_h, G_h) on each training fold without using a WGAN. The archived ablation used the direct-inversion estimate to select among roots; the revised estimator instead uses the single-crossing acceptance rule below. This experiment diagnoses the local nuisance requirement; it does not by itself establish a gain from retaining a global generative model.

A subsequent paired pilot implements a simple hybrid directly. Let n_{-k} be the training-fold size and let $\widehat{\eta}_{W,h}^{(-k)}$ and $\widehat{\eta}_{E,h}^{(-k)}$ denote the WGAN and empirical local moments. Use the same mixture weight for the CDF and density:

$$\widehat{\eta}_{C,h}^{(-k)} = (1 - w_k)\widehat{\eta}_{E,h}^{(-k)} + w_k \widehat{\eta}_{W,h}^{(-k)}, \qquad w_k = \frac{50}{n_{-k} + 50}. \tag{13}$$

The fixed pseudo-count of 50 is specified before the evaluation runs; it is not chosen using the true reserve, coverage, or evaluation-fold observations. This post-fit calibration preserves

the original WGAN-GP architecture and loss. It changes only the moments supplied to the existing orthogonal score. For fixed h, CDF signals lie in $[0,1]$ and density signals in $[0,(\sqrt{2\pi}h)^{-1}]$. Hence the difference from the empirical nuisance is uniformly $O(n^{-1})$, regardless of the WGAN fit, and the empirical-local rate result continues to apply under the same overlap and root conditions. This is an empirical-dominated safeguard, not a claim of improved asymptotic efficiency. Section 6 establishes its first-order equivalence to empirical local estimation and the same-kernel full-sample plug-in under a common single-crossing target. The protection is for fixed h and does not establish consistency of the latent distribution produced by the generator. Whether the retained WGAN component improves finite-sample performance is assessed against the empirical-only estimator and a full-sample smoothed plug-in using the same kernel and bandwidth.

As a computational precursor, I also report a fold-averaged plug-in. Partition the auctions into K folds and train $\widehat{F}^{(-k)}$ outside fold k. Define

$$\widehat{r}_{\mathrm{PI}}^{(-k)} \in \arg\max_r r\{1-\widehat{F}^{(-k)}(r)\}, \qquad \overline{r}_{\mathrm{FA}} = \frac{1}{K}\sum_{k=1}^{K}\widehat{r}_{\mathrm{PI}}^{(-k)}. \tag{14}$$

This construction measures the stability gained from sample splitting and averaging independently trained generators. Because the held-out observations do not yet enter an observation-level reserve score, $\overline{r}_{\mathrm{FA}}$ is not the orthogonal DML estimator.

6 Cross-Fitted Orthogonal Estimation

Partition the auction sample into folds $I_1,\ldots,I_K$. For each fold k, train the WGAN-GP and estimate all nuisance functions using observations outside I_k. Denote the resulting nuisance estimate by $\widehat{\eta}^{(-k)}$.

Sample splitting by itself is not sufficient for DML in this application. The fold-averaged plug-in $\overline{r}_{\mathrm{FA}}$ introduced in Section 5 uses the same training partitions but does not evaluate an observation-level score on I_k. It is therefore an ensemble benchmark. Cross-fitting acquires its inferential role only when each held-out observation enters the score below.

Let Φ and ϕ denote the standard normal CDF and density. For fixed bandwidth h, define the observable signals

$$S_h(B;\theta) = \Phi\left(\frac{\theta-B}{h}\right), \qquad K_h(B;\theta) = \frac{1}{h}\phi\left(\frac{\theta-B}{h}\right), \tag{15}$$

and the local nuisance vector $\eta_h(\theta) = (A_h(\theta), G_h(\theta))$, where $A_h(\theta) = \mathbb{E}[S_h(B;\theta)]$ and $G_h(\theta) =$

$\mathbb{E}[K_h(B;\theta)]$. The regularized reserve moment is

$$m_h(\theta, A, G) = 1 - A^{1/N} - \frac{\theta}{N} A^{1/N-1} G. \tag{16}$$

Fix a positive compact search interval $\Theta = [\underline{\theta}, \overline{\theta}] \subset (0, \infty)$ before estimation. Define $R_h(\theta) = \theta\{1 - A_h(\theta)^{1/N}\}$, so that $R_h'(\theta) = m_h(\theta, A_h(\theta), G_h(\theta))$. Under the single-crossing condition stated below, the fixed-bandwidth target $\theta_{0,h}$ is its unique interior maximizer on Θ. It need not equal the exact optimal reserve r_0: Gaussian smoothing is applied to the maximum distribution before inversion, not to the original valuation distribution. Approximation of r_0 requires a separate shrinking-bandwidth argument.

The derivatives needed for the locally robust correction are

$$m_A(\theta, A, G) = -\frac{1}{N} A^{1/N-1} - \frac{\theta}{N}\left(\frac{1}{N} - 1\right) A^{1/N-2} G, \tag{17}$$

$$m_G(\theta, A, G) = -\frac{\theta}{N} A^{1/N-1}. \tag{18}$$

The observation-level score is

$$\begin{aligned} \psi_h(B;\theta, A, G) = m_h(\theta, A, G) + m_A(\theta, A, G)\{S_h(B;\theta) - A\} \\ + m_G(\theta, A, G)\{K_h(B;\theta) - G\}. \end{aligned} \tag{19}$$

Because the two residuals have mean zero at the truth, differentiation of the score expectation adds $-m_A$ and $-m_G$ to the derivatives of the first term. Hence

$$\partial_{(A,G)} \mathbb{E}\left[\psi_h(B;\theta_{0,h}, A, G)\right]\big|_{(A,G)=\eta_{0,h}(\theta_{0,h})} = 0. \tag{20}$$

Proposition 1 (Orthogonality and remainder)**.** *At any fixed θ, let $\eta_0 = (A_h(\theta), G_h(\theta))$. Then*

$$\mathbb{E}_0 \psi_h(B;\theta,\eta) = m_h(\theta,\eta) + \nabla_\eta m_h(\theta,\eta)^\mathsf{T}(\eta_0 - \eta). \tag{21}$$

On a compact nuisance region with A bounded away from zero, this implies

$$|\mathbb{E}_0 \psi_h(B;\theta,\eta) - m_h(\theta,\eta_0)| \le C_h\{(A - A_h)^2 + |A - A_h|\,|G - G_h|\}. \tag{22}$$

If $A = A_h$, the expectation is correct for any finite G. Correct G alone does not generally have this property. The score is therefore Neyman orthogonal and locally robust, but not a two-sided doubly robust AIPW score.

For uniform nuisance errors $\delta_{A,n}$ and $\delta_{G,n}$, the population remainder is therefore negligible at the root-n scale if $\delta_{A,n}^2 + \delta_{A,n}\delta_{G,n} = o_p(n^{-1/2})$. This condition must be accompanied by

score consistency and the usual stochastic and root controls; it is not a license to substitute arbitrary quantities for either local moment.

The estimator solves

$$\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in I_k}\psi_h\Big(B_i;\widehat{\theta}_h,\widehat{A}_h^{(-k)}(\widehat{\theta}_h),\widehat{G}_h^{(-k)}(\widehat{\theta}_h)\Big)=0. \tag{23}$$

Figure 1 summarizes the calibrated implementation. The training observations supply both the WGAN and empirical local moments; the held-out observations enter only through the score signals. Scores are pooled across folds before solving for a common reserve estimate.

Write $\widehat{\Psi}_n(\theta)$ for the pooled sample score in the preceding equation. For the population moment and its derivative, use

$$m_{0,h}(\theta)=m_h(\theta,A_h(\theta),G_h(\theta)),\qquad J_h=m'_{0,h}(\theta_{0,h}).$$

The formal estimator retains an interior downward crossing only when it is the unique root on Θ, the endpoint scores have the prescribed positive and negative signs, and its total score derivative is negative and separated from zero. Bracket refinement solves the score to residual $o(n^{-1/2})$. Missing roots, multiple roots, boundary roots, active CDF clipping, or a nearly zero derivative are reported as inference failures, not replaced by a minimum-absolute-score point with a Wald interval. The implementation scans a prespecified grid and refines detected crossings; its finite-sample root count is a diagnostic, not a certificate excluding every possible hidden crossing between grid points. The theorem below uses uniform derivative convergence and a single population root to justify this rule asymptotically, without requiring such a finite-sample certificate.

Let I_k^c denote the training observations for fold k and $n_{-k}=|I_k^c|$. The empirical-local learner is

$$\widehat{A}_h^{(-k)}(\theta)=\frac{1}{n_{-k}}\sum_{j\in I_k^c}S_h(B_j;\theta),\qquad \widehat{G}_h^{(-k)}(\theta)=\frac{1}{n_{-k}}\sum_{j\in I_k^c}K_h(B_j;\theta). \tag{24}$$

Assumption 1 (Fixed-bandwidth regularity)**.** *The sampling restrictions in Section 3 hold, and $h>0$, N, and $K\geq 2$ are fixed. The following conditions apply.*

1. *The interval Θ is fixed and positive. The continuous function $m_{0,h}$ has exactly one zero $\theta_{0,h}$ in its interior, $m_{0,h}(\underline{\theta})>0$, and $m_{0,h}(\overline{\theta})<0$. On a neighborhood of that zero, $m'_{0,h}(\theta)\leq -2j_h<0$. Moreover,*

$$\inf_{\theta\in\Theta}A_h(\theta)\geq a_h>0. \tag{25}$$

Any numerical CDF clipping threshold is less than $a_h/2$.

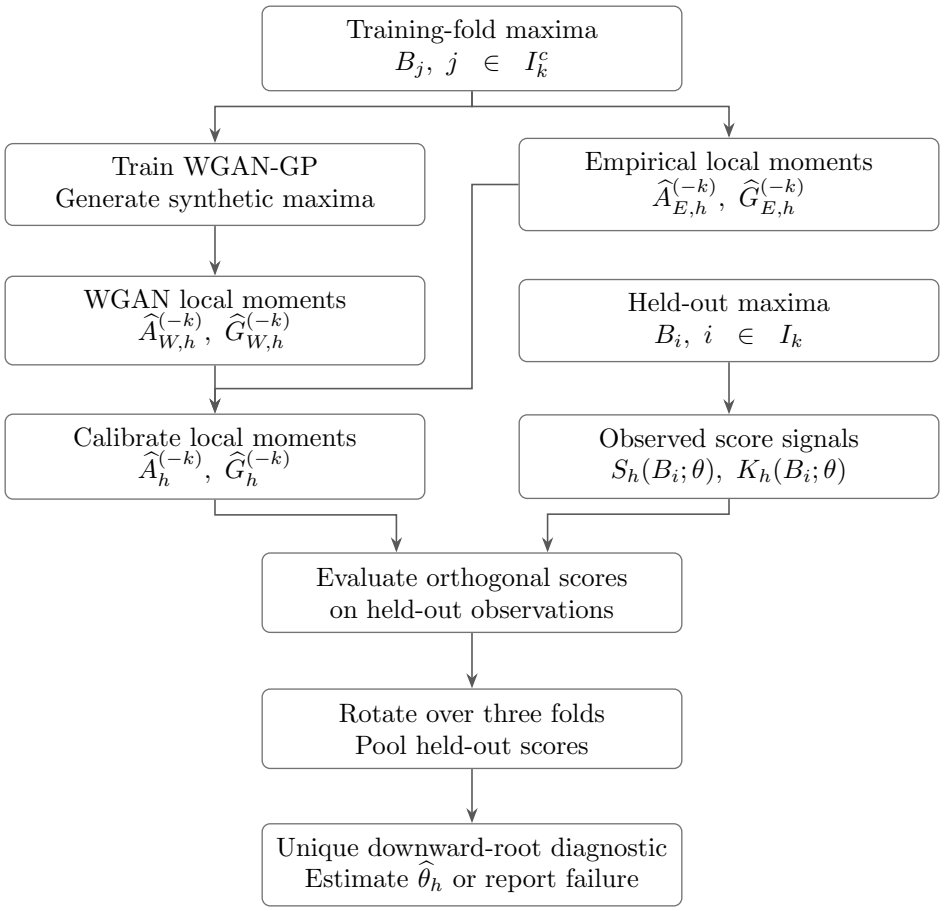


Figure 1: Cross-fitted estimation with calibrated generative local moments. Synthetic maxima are formed from N independently generated bidder values. For each candidate reserve, the training-fold WGAN and empirical moments are combined using equation (13). Training and tuning use only I_k^c. The three-fold implementation pools observation-level scores, rather than fold-specific reserve estimates. The final estimate targets the fixed-bandwidth reserve $\theta_{0,h}$ and is reported only when the prespecified root, overlap, and slope checks pass.

2. *The split is independent of the observations, and* $|I_k|/n \to \pi_k \in (0,1)$ *for each fold. Training, model selection, and synthetic sampling for a fold use no observations from its holdout set.*

3. *The nuisance is either equation* (24) *or the calibrated learner in equation* (13) *with a fixed finite pseudo-count (50 in the implementation). Synthetic maxima are finite almost surely; their number may be fixed. The same weight is used for* A_h *and* G_h *and does not depend on* θ*.*

4. *The variance* $V_h = \mathrm{Var}\{\psi_h(B;\theta_{0,h},\eta_{0,h})\}$ *is positive. The numerical routine refines its bracket to score residual* $\varepsilon_n = o(n^{-1/2})$ *and uses the total derivative of* $\widehat{\Psi}_n$*; its rejection threshold is at most* j_h*.*

These conditions state a single-crossing restriction on a specified search interval, not an assumption that an arbitrary root-selection routine is consistent. They imply that $\theta_{0,h}$ uniquely maximizes R_h on Θ. Positive variance and overlap concern the maximum distribution at this target, and need not hold uniformly over a growing number of bidders.

Proposition 2 (Primitive uniform rates)**.** *Under Assumption 1, put* $\rho_n = \sqrt{\log n/n}$*. For the empirical-local learner, and also for its fixed-pseudo-count calibration,*

$$\max_{k\le K}\sup_{\theta\in\Theta}\left\{\|\widehat{\eta}_h^{(-k)} - \eta_{0,h}\| + \|\partial_\theta\widehat{\eta}_h^{(-k)} - \partial_\theta\eta_{0,h}\|\right\} = O_p(\rho_n). \tag{26}$$

Calibration changes the empirical nuisance and its first derivative by uniformly $O(n^{-1})$*. Consequently,* $\widehat{\Psi}_n$ *and its total derivative converge uniformly to* $m_{0,h}$ *and* $m'_{0,h}$*, respectively.*

Theorem 1 (Fixed-bandwidth inference)**.** *Under Assumption 1, the probability of the stated inference failures tends to zero. On successful runs the selected root is consistent,* $\widehat{J}_h = \widehat{\Psi}'_n(\widehat{\theta}_h) \to_p J_h$*, and*

$$\sqrt{n}(\widehat{\theta}_h - \theta_{0,h}) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varphi_{i,h} + o_p(1), \qquad \varphi_{i,h} = -J_h^{-1}\psi_h(B_i;\theta_{0,h},\eta_{0,h}). \tag{27}$$

The empirical variance of the estimated influence values is consistent for V_h/J_h^2*. The usual studentized statistic is asymptotically standard normal, and its Wald interval has asymptotic coverage* $1-\gamma$ *for* $\theta_{0,h}$*, not for* r_0*.*

The total derivative includes the dependence of both nuisance functions on θ. If $T_h = (S_h, K_h)^\mathsf{T}$ and a dot denotes this derivative,

$$\frac{d\psi_h}{d\theta} = m_\theta + \nabla_\eta m_h^\mathsf{T}\dot{T}_h + \{\nabla_\eta m_\theta + \nabla_\eta^2 m_h\dot{\eta}\}^\mathsf{T}(T_h - \eta). \tag{28}$$

Here $\dot{S}_h = K_h$ and $\dot{K}_h = -(\theta - B)K_h/h^2$. A centered finite difference is also consistent if its step tends to zero, remains inside Θ, and is applied to the same smooth pooled score; the uniform derivative result supplies this conclusion. Fixed numerical tolerances in archived pilots are finite-sample approximations, not asymptotic guarantees. Failure frequencies are reported alongside coverage rather than silently treating rejected cases as estimates.

Proposition 3 (Full-sample plug-in equivalence)**.** *Let $\widetilde{\eta}_h(\theta) = n^{-1}\sum_{i=1}^n T_h(B_i;\theta)$ and let $\widetilde{\theta}_h$ be the unique downward root of $m_h(\theta, \widetilde{\eta}_h(\theta))$ on Θ, refined to the same residual tolerance. Under Assumption 1, it has the influence function in equation* (27)*, and*

$$\sqrt{n}(\widehat{\theta}_h - \widetilde{\theta}_h) \to_p 0. \tag{29}$$

Thus empirical-local DML, fixed-pseudo-count calibrated DML, and the same-kernel full-sample plug-in have the same first-order asymptotic variance.

This equivalence does not rule out finite-sample differences. It does rule out attributing a first-order variance improvement to the WGAN component under these assumptions. The proof uses the empirical part of the calibrated learner and makes no consistency claim for the standalone WGAN. For a pure WGAN score, local function and derivative consistency, a negligible remainder in equation (22), and control of synthetic-sample error require separate verification. A low training critic objective does not supply those conditions. At fixed h, a rate for the true Wasserstein distance can control the Lipschitz kernel moments; their constants deteriorate as h shrinks, and a fitted critic value is not such a population rate.

6.1 Confidence sets and conservative envelopes

The Wald interval in Theorem 1 uses the score slope and variance evaluated at the estimated root. A second construction retains the nonlinear score curve and evaluates its variance at each candidate reserve. For $i \in I_k$, write

$$\widehat{\psi}_{i,h}(\theta) = \psi_h\Big(B_i; \theta, \widehat{\eta}_h^{(-k)}(\theta)\Big), \qquad \widehat{V}_n(\theta) = \frac{1}{n-1}\sum_{i=1}^n \{\widehat{\psi}_{i,h}(\theta) - \widehat{\Psi}_n(\theta)\}^2.$$

This is the sample variance of the pooled observation-level scores, not the variance of the K fold means. Define the studentized score and its continuous acceptance set by

$$T_n(\theta) = \frac{\sqrt{n}\,\widehat{\Psi}_n(\theta)}{\sqrt{\widehat{V}_n(\theta)}}, \qquad \mathcal{C}_{n,1-\gamma} = \{\theta \in \Theta : |T_n(\theta)| \le z_{1-\gamma/2}\}. \tag{30}$$

The set is defined at candidates where the statistic is well defined. On a failed point-estimation run no confidence set is reported; for coverage accounting the set is taken to be empty. The critical value is fixed before the experiment; it is 1.96 in the 95 percent comparisons below. Neither the point estimate nor the nuisance-learning procedure is changed.

Proposition 4 (Fixed-bandwidth score-set coverage)**.** *Under Assumption 1,* $T_n(\theta_{0,h}) \Rightarrow \mathcal{N}(0,1)$*, and the continuous set in equation* (30) *satisfies*

$$\Pr\{\theta_{0,h} \in \mathcal{C}_{n,1-\gamma}\} \longrightarrow 1-\gamma.$$

If the set is nonempty, define its conservative interval envelope as

$$\mathcal{E}_{n,1-\gamma} = [\inf \mathcal{C}_{n,1-\gamma}, \sup \mathcal{C}_{n,1-\gamma}], \tag{31}$$

and set $\mathcal{E}_{n,1-\gamma} = \varnothing$ *when* $\mathcal{C}_{n,1-\gamma} = \varnothing$*. Then* $\mathcal{C}_{n,1-\gamma} \subseteq \mathcal{E}_{n,1-\gamma}$ *for every sample, and*

$$\liminf_{n\to\infty} \Pr\{\theta_{0,h} \in \mathcal{E}_{n,1-\gamma}\} \geq 1-\gamma.$$

These claims concern the fixed-bandwidth target and the mathematical sets, not exact finite-sample coverage or a certificate for a finite search grid.

The proof in Appendix A.4 uses the same orthogonal remainder and variance consistency as the Wald result. Inverting pointwise tests does not require a simultaneous confidence band over Θ: coverage is the nonrejection event at the one true parameter. The argument does not establish robustness to weak identification, growing bidder counts, or shrinking bandwidths.

The full acceptance set can be disconnected even when the estimated score has a unique zero. Candidate-dependent studentization may produce a distant accepted region where the estimated variance is large. I retain every detected component, rather than selecting the component containing $\widehat{\theta}_h$. Such selection could remove covered parameter values. The interval envelope fills the rejected gaps between components. It is therefore reported separately and is not called the original score-test confidence set or an improvement in estimation precision.

The implementation evaluates 281 equally spaced candidates on Θ, adds the estimated root, and refines detected acceptance-boundary crossings by Brent's method with tolerance 10^{-11}. It records the components without joining gaps. Separate flags identify more than one component and contact with either search boundary, using tolerance 10^{-9} only for the flag. The boundaries are not moved by this tolerance. A boundary flag refers to the prespecified

search interval, not to a data-identified support endpoint. Nonfinite candidate moments or nonpositive score variance are recorded as inference failures; they are not silently dropped from the search. In contrast, disconnection and boundary contact are reporting flags, not computational failures. Failure rates use all replications, while reported lengths condition on a successfully constructed set. For a disconnected set, its length is the sum of component lengths; the envelope length includes the intervening gaps. Finer grids check numerical sensitivity but cannot certify the absence of every arbitrarily narrow component.

6.2 Diagnostic extensions

To diagnose a failure of the WGAN-based nuisance, I also evaluate two versions of the identical score. The empirical-local version replaces generated maxima with observed maxima from the corresponding training fold when estimating (A_h, G_h). The oracle version supplies their population values from the known Monte Carlo DGP. Agreement between these two versions, together with failure of the WGAN version, isolates the problem in nuisance approximation rather than in equation (19).

Inference for the exact reserve adds a separate bandwidth problem. The fixed-bandwidth theorem does not supply a triangular-array result. Under local smoothness, Gaussian regularization produces a leading approximation bias of order h^2, while the density component of the score has stochastic scale of order $(nh)^{-1/2}$. A heuristic undersmoothing requirement is therefore $nh^5 \to 0$, together with $nh \to \infty$. For a power rule $h_n \propto n^{-\alpha}$, this suggests $1/5 < \alpha < 1$. These rate inequalities do not guarantee stable finite-sample root selection, especially because the maximum- bid CDF is small near the reserve. The simulation below reports both coverage of θ_{0,h_n} and coverage of r_0 rather than treating them as the same target.

To remove the leading smoothing bias without forcing h_n into the unstable region, suppose the selected population root admits the local expansion

$$\theta_{0,h} = r_0 + b_2 h^2 + O(h^4). \tag{32}$$

For a fixed ratio $\lambda > 1$, define the Richardson estimator

$$\widehat{r}_{\mathrm{R}} = \frac{\lambda^2 \widehat{\theta}_h - \widehat{\theta}_{\lambda h}}{\lambda^2 - 1}. \tag{33}$$

The $O(h^2)$ term cancels in its population analogue. Both estimates use the same folds and observations, so their dependence must enter inference. If $\widehat{\varphi}_{i,h}$ and $\widehat{\varphi}_{i,\lambda h}$ are their estimated

influence values, the corrected influence value is

$$\widehat{\varphi}_{i,\mathrm{R}} = \frac{\lambda^2 \widehat{\varphi}_{i,h} - \widehat{\varphi}_{i,\lambda h}}{\lambda^2 - 1}, \qquad \widehat{\mathrm{se}}(\widehat{r}_{\mathrm{R}}) = \frac{\widehat{\mathrm{sd}}(\widehat{\varphi}_{i,\mathrm{R}})}{\sqrt{n}}. \tag{34}$$

The implementation sets $\lambda = \sqrt{2}$, so $\widehat{r}_{\mathrm{R}} = 2\widehat{\theta}_h - \widehat{\theta}_{\sqrt{2}h}$. It selects the smoother root first and uses it as the reference for the smaller-bandwidth root. This continuation rule reduces arbitrary root switching but does not eliminate multiple-root uncertainty.

The following extension is conditional: the joint linear representation, higher-order bias expansion, and covariance conditions are additional assumptions, not consequences of Theorem 1. In particular, the differentiability imposed on F_0 in Section 3 is not enough by itself to justify the fourth-order expansion.

Proposition 5 (Conditional exact-reserve Richardson inference)**.** *Let $h_n \to 0$ and fix $\lambda > 1$. Suppose the roots at h_n and λh_n have a joint linear representation with remainder $o_p(s_{\mathrm{R},n}/\sqrt{n})$ after the Richardson combination and a valid triangular-array central limit theorem. Suppose their covariance estimator is consistent, the relevant population roots are locally unique and separated, and*

$$\theta_{0,h} = r_0 + b_2 h^2 + b_4 h^4 + o(h^4). \tag{35}$$

If $s^2_{\mathrm{R},n}$ is the variance of the Richardson influence value and

$$\frac{\sqrt{n} h_n^4}{s_{\mathrm{R},n}} \to 0, \tag{36}$$

then the covariance-aware Richardson statistic, centered at r_0 and divided by $s_{\mathrm{R},n}/\sqrt{n}$, converges to a standard normal distribution. When $s_{\mathrm{R},n} \asymp h_n^{-1/2}$, a sufficient bandwidth condition is $nh_n \to \infty$ and $nh_n^9 \to 0$.

Proposition 5 isolates what the finite-sample experiment must check: local nuisance rates along the bandwidth path, root separation, covariance across bandwidths, and residual bias relative to the standard error. Proofs are in the appendix.

The remaining $O(h^4)$ term can be diagnosed with a third bandwidth. Strengthen the local expansion to

$$\theta_{0,h} = r_0 + b_2 h^2 + b_4 h^4 + O(h^6) \tag{37}$$

and let $\widehat{r}_{\mathrm{R}}(h)$ denote equation (33). The adjacent extrapolation is $\widehat{r}_{\mathrm{R}}(\lambda h)$, constructed from λh and $\lambda^2 h$. Since

$$r_{\mathrm{R}}(\lambda h) - r_{\mathrm{R}}(h) = (\lambda^4 - 1)\{r_{\mathrm{R}}(h) - r_0\} + O(h^6), \tag{38}$$

the absolute residual-bias diagnostic is

$$\widehat{B}_h = \frac{|\widehat{D}_h| + 1.96\,\widehat{\mathrm{se}}(\widehat{D}_h)}{\lambda^4 - 1}, \qquad \widehat{D}_h = \widehat{r}_{\mathrm{R}}(\lambda h) - \widehat{r}_{\mathrm{R}}(h). \tag{39}$$

The standard error of $\widehat{D}_h$ again combines observation-level influence values across all three bandwidths and therefore retains their covariance. I report the bias-envelope interval

$$\left[\widehat{r}_{\mathrm{R}}(h) - 1.96\widehat{\mathrm{se}}_{\mathrm{R}} - \widehat{B}_h,\ \widehat{r}_{\mathrm{R}}(h) + 1.96\widehat{\mathrm{se}}_{\mathrm{R}} + \widehat{B}_h\right]. \tag{40}$$

For root robustness, I enumerate every grid crossing at h, connect each to the nearest root at λh and then $\lambda^2 h$, construct equation (40) for every continuation chain, and take the union of those intervals. This retains disconnected components rather than silently selecting one root. The procedure is a finite-sample diagnostic, not an honest uniform confidence set: its interpretation requires a stable even- order bias expansion, adequate grid coverage, and reliable local Gaussian approximations along each root chain.

7 Monte Carlo Evidence

The primary comparison below uses a fresh validation sample after the fixed-bandwidth numerical safeguards and score-inversion rule were fixed. The subsequent historical experiments are retained as diagnostics and are explicitly distinguished from this validation. The revised code rejects absent, multiple, boundary, weak-slope, or clipping-affected roots instead of returning a minimum-residual grid point. Every failure remains in the coverage denominator. Lengths and conditional error summaries identify their valid-run sample. A disconnected or boundary-touching confidence set is a separate diagnostic flag, not a numerical failure.

7.1 Fresh validation of fixed-bandwidth confidence sets

Values satisfy $\log V_{ij} \sim \mathcal{N}(0, 0.5^2)$ with $N = 5$ bidders. The econometrician observes the maximum in each auction. I generate 100 independent repetitions at each of $n = 500$ and $n = 2000$, with base seed 2026110800. These data, folds, GAN training seeds and synthetic samples do not overlap the earlier development experiment. Each repetition shares data across calibrated WGAN-DML, empirical-local DML and the same-kernel full-sample plug-in; the two DML estimators also share the three-fold split. The bandwidth is $h = 0.15$, the pseudo-count is 50, each GAN receives 1,200 generator updates and five critic updates per generator update, and each training fold supplies 50,000 generated maxima. None of these settings is retuned in the validation.

All intervals and sets target $\theta_{0,h} = 0.831828\ldots$, not the exact unsmoothed reserve. For the same point estimate, I compare the original 95 percent Wald interval with the full score-inversion set in Section 6.1. The latter uses a 281-point grid augmented by the estimated root, with crossing endpoints refined to tolerance 10^{-11}. The prespecified search interval remains $[0.45, 1.15]$. Repetitions 1, 50 and 100 also use a 561-point grid as a numerical check. All detected components are retained. Coverage of the full set is checked against the test at the known simulation target. The conservative envelope is a separate output, not a new point estimator or a replacement for the accepted set. This reporting layer was adopted after examining the disconnected validation cases. It leaves their original accepted values and all Wald–score comparisons unchanged; it is not a separately preregistered inferential improvement.

Table 1: Independent validation of Wald intervals and full score sets

		Coverage (%)		Mean length			Counts
n	Method	Wald	Full set	Wald	Full set	Envelope	F/D/B
500	Calibrated GAN	91	93	0.14215	0.14614	0.14917	0/2/2
500	Empirical DML	91	93	0.13911	0.14208	0.14616	0/2/2
500	Smooth plug-in	91	92	0.12364	0.12303	0.12303	0/0/0
2,000	Calibrated GAN	96	95	0.06727	0.06741	0.06741	0/0/0
2,000	Empirical DML	96	95	0.06665	0.06679	0.06679	0/0/0
2,000	Smooth plug-in	95	95	0.06418	0.06426	0.06426	0/0/0

Notes: Each row uses 100 independent repetitions and the fixed target $\theta_{0,h}$. Coverage is in percent, with failures counted as noncoverage. Lengths are averaged over valid outputs. Full-set length sums component lengths and excludes rejected gaps; envelope length includes those gaps. F/D/B reports computational failures, disconnected full sets and full sets touching a search boundary. Both interval methods have zero computational failures in these cells. Envelope reporting retains the source-set flags.

At $n = 500$, score inversion raises coverage from 91 to 93 percent for both calibrated GAN and empirical-local DML, and from 91 to 92 percent for the plug-in. For calibrated GAN, two repetitions are gained and none lost relative to Wald. The paired Monte Carlo standard error of the two-percentage-point difference is 1.41 percentage points; the exact two-sided test based on the two discordant pairs has $p = 0.50$. The Wilson 95 percent interval for the 93 percent coverage estimate is approximately $[86.25, 96.57]$ percent. The direction of the development-sample gain is reproduced, but its magnitude is imprecise and does not establish uniformly better coverage or a repair to the nominal level. At $n = 2000$,

full-set coverage is 95 percent in all three methods, compared with 96, 96 and 95 percent under Wald. These differences do not support a coverage ranking at the larger sample size.

Two replications, 29 and 50, produce disconnected sets for both calibrated GAN and empirical-local DML. Each has an additional low-end component touching 0.45; the plug-in sets are connected. This raises the distinction between a confidence *set* and an interval envelope. For calibrated GAN at $n = 500$, mean full-set length is 0.14614 and mean envelope length is 0.14917, versus 0.14215 for Wald. The corresponding empirical-local lengths are 0.14208 and 0.14616, versus 0.13911. Envelope coverage happens to equal full-set coverage in this validation. Its set-inclusion guarantee is not evidence of improved estimation accuracy.

The appendix examines these cases without changing the validation outputs. Dense-grid and variance-aggregation checks reproduce the extra components, and the low-end variance is dominated by one held-out observation. Fixing the variance at the estimated root eliminates the disconnected sets in an exploratory replay but reduces calibrated-GAN and empirical-local coverage from 93 to 90 percent at $n = 500$ (Table 15). It is therefore not adopted. The reporting rule keeps the complete score set, flags disconnected and boundary-touching cases, and supplies its conservative envelope when a single interval is needed. No component is deleted after observing its location or its contribution to coverage.

7.2 Scope of the archived experiments

The remaining tables and figures in this section predate the revised root-acceptance safeguards. They are not pooled with the validation above. The direct-inversion benchmark now uses the empirical CDF left limit at a candidate price, so that bids equal to the reserve are eligible to trade. The archived direct-inversion numbers below used right-continuous empirical CDF values at observed prices. New runs are saved separately and do not overwrite these archived results.

7.3 Direct-inversion benchmark

The first experiment fixes the baseline model before introducing WGAN-GP or DML. Values are lognormal with $\log V_{ij} \sim \mathcal{N}(0, 0.5^2)$, every auction has $N = 5$ bidders, and the econometrician observes only the maximum. For each sample size $n \in \{200, 500, 2000, 10000\}$, I generate 500 Monte Carlo samples. The random seed is fixed at 20260716. In every replication, $\widehat{F}_{\mathrm{DI}} = \widehat{H}_n^{1/N}$ and the reserve maximizes $r\{1 - \widehat{F}_{\mathrm{DI}}(r)\}$ over the observed maxima.

This experiment supplies a numerical identification check and a calibration target for

Table 2: Direct-inversion reserve estimates

n	Mean $\widehat{r}$	Bias	SD	RMSE	Mean regret
200	0.9053	0.1335	0.0794	0.1552	0.0032
500	0.8446	0.0728	0.0635	0.0965	0.0009
2000	0.7960	0.0242	0.0556	0.0606	0.0002
10000	0.7584	-0.0134	0.0513	0.0530	0.0001

Notes: The true reserve solves $1 - F_0(r) - r f_0(r) = 0$. Revenue regret is $\mathcal{R}_N(r_0; F_0) - \mathcal{R}_N(\widehat{r}; F_0)$.

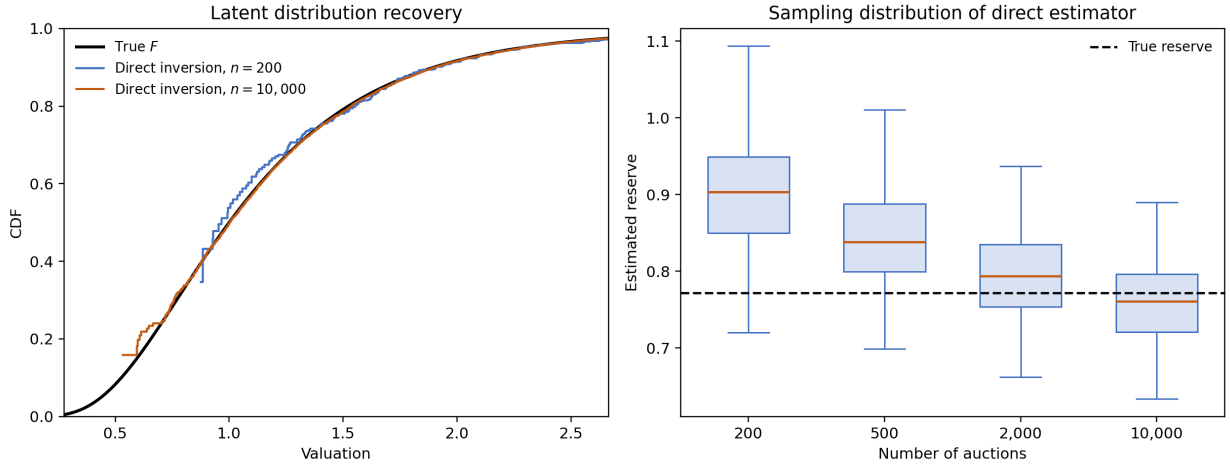


Figure 2: Recovery of the latent CDF and sampling distribution of the direct-inversion reserve estimator

every later learner. The left panel of Figure 2 shows where the inversion has little information; the right panel and Table 2 measure whether this translates into economically material reserve error.

7.4 Structural WGAN-GP pilot

I next replace direct inversion with the structural generator described in Section 5 while retaining the same lognormal DGP and $N = 5$. The pilot uses five training seeds at each $n \in \{500, 2000, 10000\}$. Each run has 1,200 generator updates, five critic updates per generator update, a batch size of 512, and a gradient-penalty weight of 0.1. The generator checkpoint minimizes the observable maximum-bid Wasserstein distance. These five repetitions diagnose the pipeline and its failure modes; the final Monte Carlo will use substantially more repetitions.

Table 3: Structural WGAN-GP pilot estimates

n	Estimator	Mean $\widehat{r}$	Bias	RMSE	Mean regret	Max-bid W_1
500	Direct	0.8245	0.0527	0.0757	0.0004	–
500	WGAN-GP	0.7301	-0.0417	0.0687	0.0001	0.0390
2000	Direct	0.7962	0.0244	0.0394	0.0001	–
2000	WGAN-GP	0.7298	-0.0421	0.0579	0.0001	0.0337
10000	Direct	0.7596	-0.0123	0.0593	0.0001	–
10000	WGAN-GP	0.7247	-0.0472	0.0612	0.0001	0.0391

Notes: Five training seeds per sample size. Max-bid W_1 compares the observed and generated auction maxima and is used only for WGAN-GP checkpoint selection. The true reserve is 0.7719 and is not supplied to either estimator.

Table 3 shows that the WGAN-GP closely matches the observable maximum distribution, with mean W_1 between 0.034 and 0.039, but retains a reserve-price bias of roughly -0.04 in this pilot. The left panel of Figure 3 explains the discrepancy: the learned latent CDF is close over much of its support but misses part of the lower tail that is informative about the reserve. At $n = 500$, WGAN-GP reduces RMSE relative to direct inversion from 0.0757 to 0.0687; at the larger pilot sample sizes it does not improve RMSE. This is evidence that observable distributional fit is not sufficient for accurate scalar estimation. It motivates the next cross-fitting and orthogonal-score experiments, without presuming that those steps will eliminate the bias.

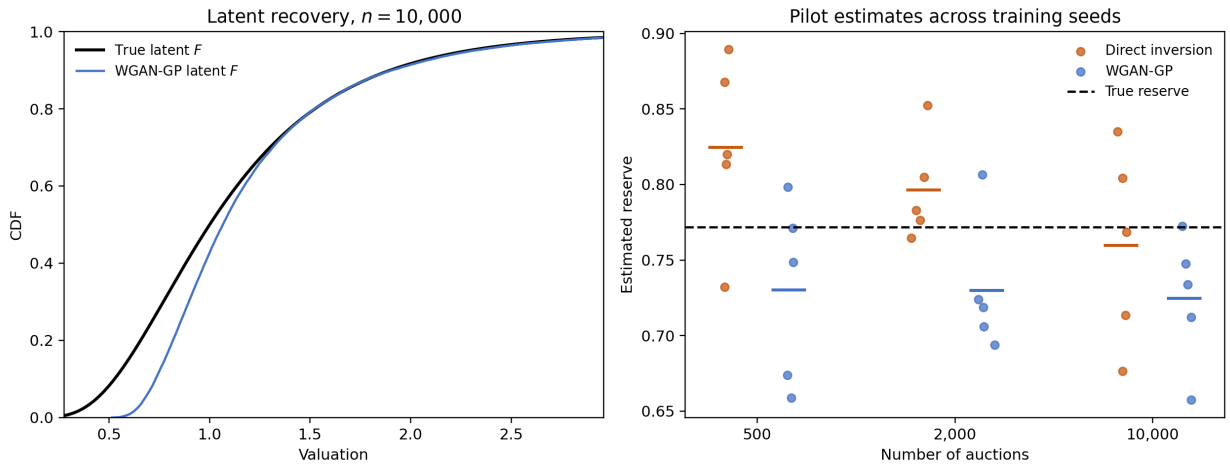


Figure 3: Latent-distribution recovery and reserve estimates in the WGAN-GP pilot

7.5 Fold-averaged WGAN precursor

I next partition each Monte Carlo sample into three folds. For each fold, the structural WGAN-GP is trained on the other two folds using the same architecture and 1,200-step training convention. The plug-in reserves from the three training samples are averaged. Held-out auctions are used only to calculate an out-of-fold maximum-bid Wasserstein diagnostic. Thus, this experiment isolates sample splitting and generator averaging from the orthogonal-score correction.

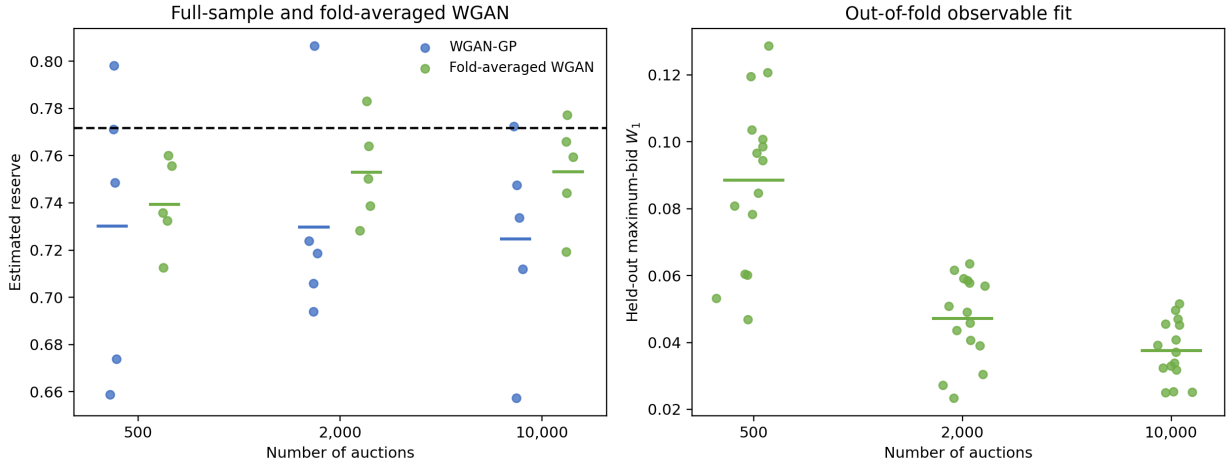


Figure 4: Reserve stability and out-of-fold observable fit under three-fold WGAN training

Table 4 shows that fold averaging reduces WGAN RMSE from 0.0687 to 0.0368 at $n = 500$, from 0.0579 to 0.0270 at $n = 2000$, and from 0.0612 to 0.0274 at $n = 10000$. The gain

Table 4: Three-fold WGAN plug-in precursor

n	Estimator	Mean $\widehat{r}$	Bias	RMSE	Mean regret	Held-out W_1	Fold SD
500	Direct	0.8245	0.0527	0.0757	0.0004	–	–
500	Full WGAN	0.7301	-0.0417	0.0687	0.0001	–	–
500	Fold average	0.7393	-0.0325	0.0368	0.0000	0.0885	0.0346
2000	Direct	0.7962	0.0244	0.0394	0.0001	–	–
2000	Full WGAN	0.7298	-0.0421	0.0579	0.0001	–	–
2000	Fold average	0.7529	-0.0190	0.0270	0.0000	0.0472	0.0275
10000	Direct	0.7596	-0.0123	0.0593	0.0001	–	–
10000	Full WGAN	0.7247	-0.0472	0.0612	0.0001	–	–
10000	Fold average	0.7532	-0.0186	0.0274	0.0000	0.0375	0.0418

Notes: Five common data seeds per sample size. "Fold average" is the mean of three plug-in reserves, each obtained from a WGAN trained outside one fold. Held-out W_1 compares generated maxima with the excluded auctions. Fold SD is the within-replication standard deviation of the three reserve estimates. This estimator is not yet Neyman orthogonal.

is mainly lower dispersion across data and training seeds. Bias remains negative, between -0.033 and -0.019. Figure 4 also shows that held-out maximum-bid W_1 falls with sample size, while the reserve does not converge monotonically in this small pilot. These results support using cross-fitting as the nuisance training architecture, but they do not attribute the improvement to DML or claim that sample splitting removes first-order nuisance bias.

7.6 Observation-level orthogonal-score pilot

I next implement equation (19) using the same three folds, WGAN architecture, data seeds, and 1,200-step training convention. The Gaussian bandwidth is fixed at $h = 0.15$, and each training-fold generator produces 50,000 synthetic maxima for the nuisance moments. Held-out auctions enter the score through both the smoothed-CDF and kernel-density residuals. The true fixed-bandwidth target is $\theta_{0,h} = 0.8318$, compared with the exact reserve $r_0 = 0.7719$; their difference is regularization bias, not estimation error.

Table 5 illustrates that formal orthogonality alone does not repair the current WGAN nuisance fit. The orthogonal correction raises the fold-averaged estimate by 0.123, 0.122, and 0.145 on average as n increases. Relative to r_0, its RMSE is 0.1355, 0.1046, and 0.1276, compared with 0.0368, 0.0270, and 0.0274 for fold averaging. Coverage of r_h is 40 percent, 60 percent, and zero. The intervals narrow with n even though the bias does not, producing

Table 5: Observation-level orthogonal WGAN-DML pilot

n	Estimator	Mean $\widehat{r}$	Bias to r_0	RMSE to r_0	RMSE to r_h	Mean SE	Coverage of r_h
500	Direct	0.8245	0.0527	0.0757	0.0549	–	–
500	Full WGAN	0.7301	-0.0417	0.0687	0.1154	–	–
500	Fold average	0.7393	-0.0325	0.0368	0.0941	–	–
500	Orthogonal DML	0.8623	0.0905	0.1355	0.1054	0.0374	0.4000
2000	Direct	0.7962	0.0244	0.0394	0.0472	–	–
2000	Full WGAN	0.7298	-0.0421	0.0579	0.1095	–	–
2000	Fold average	0.7529	-0.0190	0.0270	0.0812	–	–
2000	Orthogonal DML	0.8749	0.1030	0.1046	0.0468	0.0269	0.6000
10000	Direct	0.7596	-0.0123	0.0593	0.0926	–	–
10000	Full WGAN	0.7247	-0.0472	0.0612	0.1140	–	–
10000	Fold average	0.7532	-0.0186	0.0274	0.0811	–	–
10000	Orthogonal DML	0.8982	0.1263	0.1276	0.0688	0.0115	0.0000

Notes: Five common data seeds per sample size. RMSE is reported relative to both the exact reserve r_0 and fixed-bandwidth target r_h. Mean SE and coverage apply only to the orthogonal estimator. With five repetitions, coverage is diagnostic rather than a precise Monte Carlo estimate.

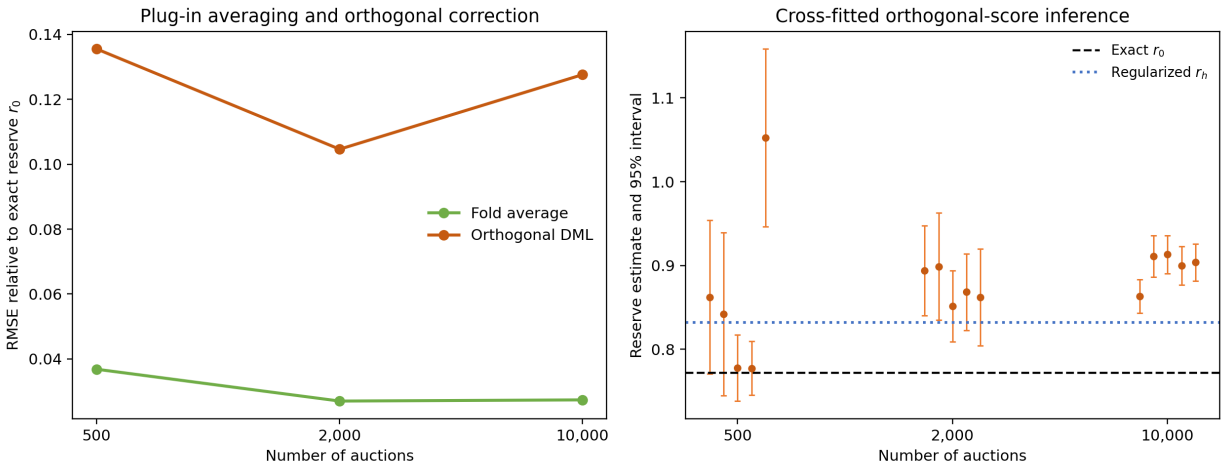


Figure 5: Finite-sample behavior of the orthogonal WGAN-DML correction

especially severe undercoverage at $n = 10000$.

This failure is informative. Orthogonality removes the first-order derivative with respect to (A_h, G_h) at the truth; it does not make a large local nuisance error small. In this design, r_0 lies in the extreme lower tail of the maximum-bid distribution, and the global WGAN fit does not deliver the local rate needed to control the second-order remainder. Figure 5 therefore motivates local tail-weighting, additional observed order statistics, or a stronger structural restriction before using the influence-function standard errors for inference.

7.7 Oracle and empirical-local nuisance ablation

I next hold the score and bandwidth fixed and change only its nuisance inputs. The empirical-local estimator calculates (A_h, G_h) directly from maximum bids in each training fold. The oracle supplies their population values from the known DGP. Both designs use 200 repetitions per sample size; the WGAN column retains the five costly training seeds above.

Table 6: Local-nuisance ablation for the orthogonal reserve score

n	Score nuisance	Reps	Mean $\widehat{r}_h$	Bias to r_h	RMSE to r_h	Mean SE	Coverage
500	WGAN generated	5	0.8623	0.0305	0.1054	0.0374	0.4000
500	Training-fold empirical	200	0.8314	-0.0005	0.0350	0.0335	0.9000
500	Oracle	200	0.8272	-0.0046	0.0311	0.0304	0.9400
2000	WGAN generated	5	0.8749	0.0430	0.0468	0.0269	0.6000
2000	Training-fold empirical	200	0.8344	0.0026	0.0171	0.0172	0.9450
2000	Oracle	200	0.8329	0.0011	0.0164	0.0167	0.9600
10000	WGAN generated	5	0.8982	0.0664	0.0688	0.0115	0.0000
10000	Training-fold empirical	200	0.8334	0.0016	0.0079	0.0076	0.9500
10000	Oracle	200	0.8332	0.0014	0.0078	0.0075	0.9550

Notes: The target is $r_h = 0.8318$ and $h = 0.15$. WGAN results use five repetitions; training-fold empirical and oracle results use 200. Coverage is for nominal 95 percent influence-function intervals for r_h.

Table 6 shows that the empirical-local score nearly reproduces the oracle. Its coverage is 90.0, 94.5, and 95.0 percent as n increases, compared with 94.0, 96.0, and 95.5 percent under the oracle. Its RMSE relative to r_h is 0.0350, 0.0171, and 0.0079; the oracle values are 0.0311, 0.0164, and 0.0078. Every replication has a score root on the search grid. Figure 6 contrasts this convergence with the persistent WGAN-nuisance error.

The ablation also quantifies the information problem. The sample contains on average only 1.05, 5.04, and 25.69 observed maxima at or below the exact reserve for $n = 500, 2000, 10000$. Nevertheless, direct training-fold moments become accurate enough under

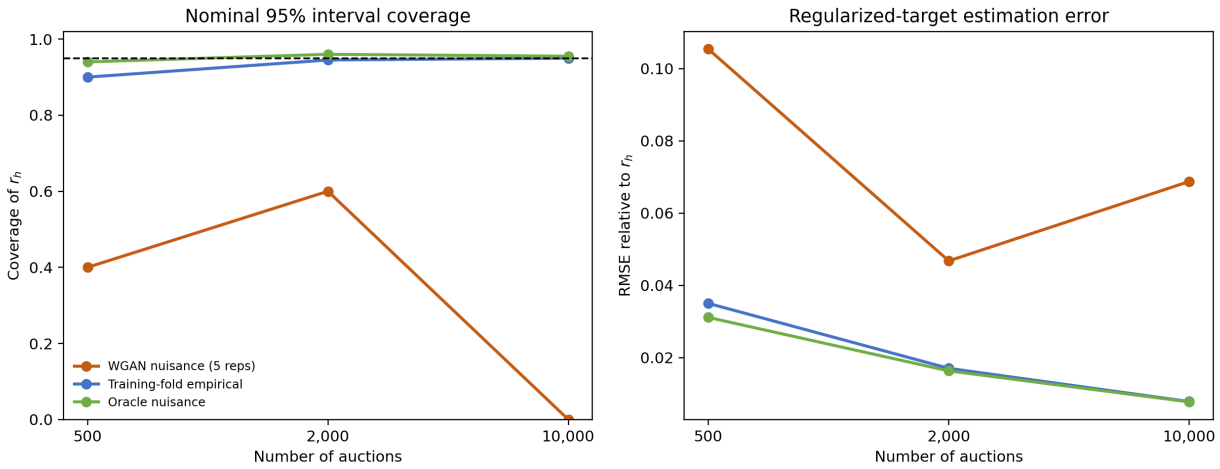


Figure 6: Coverage and RMSE under alternative local score nuisances

the fixed bandwidth. Thus the earlier failure is not a failure of the orthogonal algebra. It is evidence that a globally trained WGAN is an unnecessarily indirect and currently inaccurate estimator of observable local score moments. These results motivate a workflow using WGAN-GP for the full latent distribution and structural counterfactuals, with dedicated training-fold local moments supplying reserve inference. The empirical-only ablation does not measure a gain from WGAN participation in the score.

7.8 A simple paired local calibration

Equation (13) implements a small post-fit change to WGAN-DML. The architecture, adversarial loss, three folds, 1,200 training steps, and 50,000 generated maxima per fold are unchanged. A fixed pseudo-count of 50 gives the WGAN about 13.0, 3.6, and 0.74 percent weight as n increases. The comparison uses ten new repetitions per sample size, with base seed 20260928, following a one-sample development check with seed 20260927. All variants share the data, folds, and trained generators. Every repetition is retained; no estimator uses the known target in training or selection.

In addition to original and calibrated WGAN-DML, I report empirical-local DML and a full-sample smoothed plug-in with the identical Gaussian kernel and bandwidth. The archived pilot selected roots using the same direct-inversion reference and refined grid brackets to tolerance 10^{-10}. These numerical results have not been regenerated under the revised single-crossing and failure-reporting rules of Section 6. Future runs use the revised implementation and preserve failed replications explicitly. The fixed-bandwidth panel targets r_h

at $h = 0.15$. The exact-reserve panel applies the existing covariance-aware Richardson combination on the $n^{-0.2}$ bandwidth path; this remains a diagnostic rather than an exact-reserve coverage theorem.

Table 7: Paired local calibration: fixed-bandwidth target

n	Method	Bias	RMSE	Coverage	CI length
500	WGAN-DML	0.0572	0.0713	80.0%	0.1780
500	Calibrated WGAN	0.0258	0.0456	100.0%	0.1416
500	Empirical-local	0.0249	0.0452	100.0%	0.1377
500	Smooth plug-in	0.0139	0.0357	100.0%	0.1230
2,000	WGAN-DML	0.0471	0.0654	50.0%	0.0896
2,000	Calibrated WGAN	0.0044	0.0182	90.0%	0.0686
2,000	Empirical-local	0.0043	0.0182	90.0%	0.0680
2,000	Smooth plug-in	0.0019	0.0162	90.0%	0.0656
10,000	WGAN-DML	0.0370	0.0444	30.0%	0.0420
10,000	Calibrated WGAN	-0.0019	0.0080	100.0%	0.0294
10,000	Empirical-local	-0.0019	0.0080	100.0%	0.0293
10,000	Smooth plug-in	-0.0024	0.0077	100.0%	0.0290

Notes: Ten repetitions per cell, $N = 5$, $h = 0.15$, and $r_h = 0.8318$. Bias, RMSE, coverage, and mean 95 percent interval length refer to r_h, not $r_0 = 0.7719$. The pseudo-count is fixed in advance. With ten repetitions, coverage moves in ten-percentage-point increments; Wilson intervals in Figure 7 display this uncertainty.

Table 7 shows that the calibration reduces fixed-bandwidth RMSE from 0.0713, 0.0654, and 0.0444 to 0.0456, 0.0182, and 0.0080, respectively. Coverage changes from 80, 50, and 30 percent to 100, 90, and 100 percent while mean interval length falls in every case. These are pilot estimates: even ten successes out of ten give a Wilson lower bound of only 72.2 percent. They should not be read as evidence of exact or conservative coverage in repeated use.

The empirical-local estimates are nearly identical to the calibrated ones. The same-kernel smoothed plug-in has still lower fixed-bandwidth RMSE at all three sample sizes. Thus the

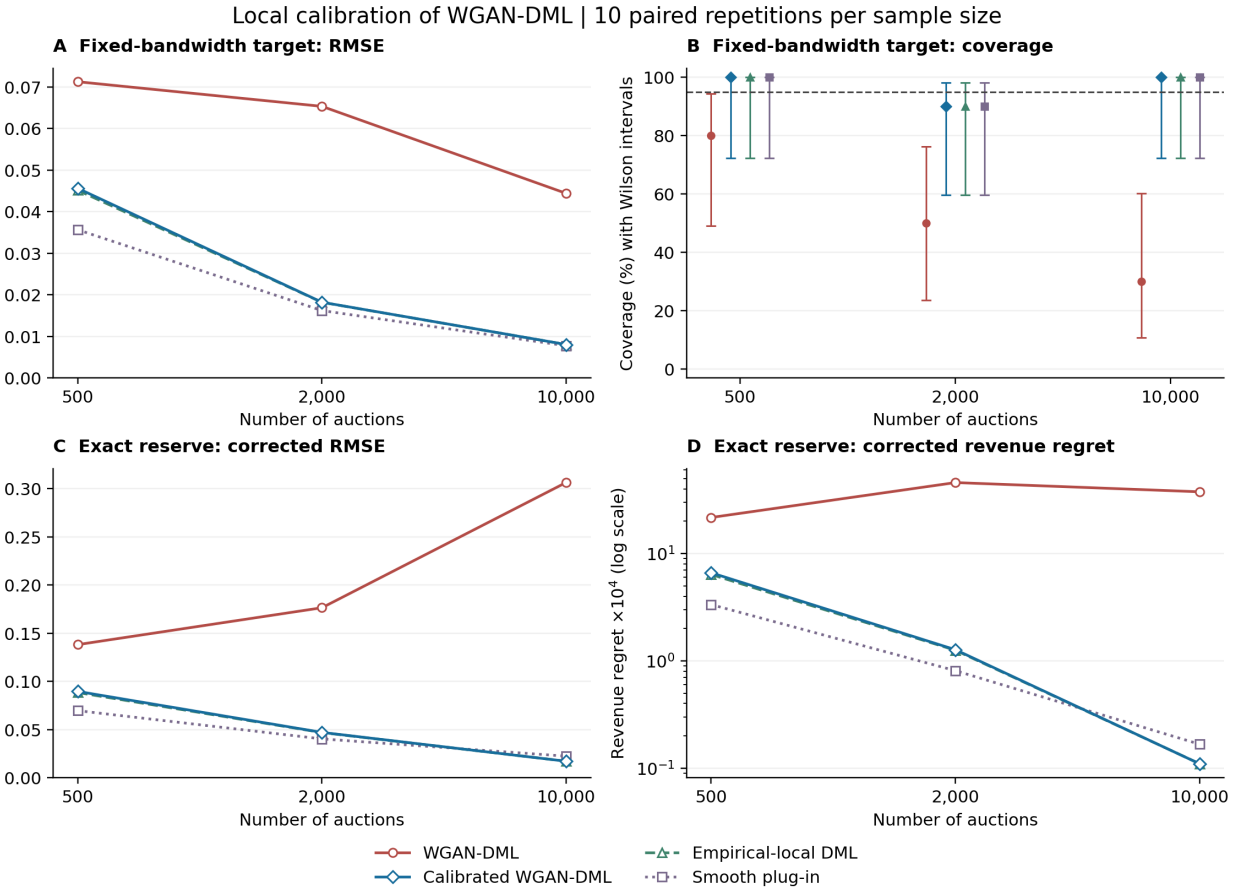


Figure 7: Paired performance after a simple local calibration. Panels A–B concern the fixed-bandwidth target; C–D concern the Richardson-corrected exact-reserve diagnostic. Revenue regret is shown on a logarithmic scale. Error bars in B are 95 percent Wilson intervals for Monte Carlo coverage.

improvement over raw WGAN-DML establishes a useful safeguard against its local error, but does not establish incremental accuracy from retaining the WGAN. Paired squared-error differences and their Monte Carlo standard errors are included in the replication archive.

For the exact-reserve diagnostic, calibrated Richardson RMSE is 0.0897, 0.0472, and 0.0172. Its observed coverage is 80, 90, and 100 percent, again with substantial Monte Carlo uncertainty. Uncalibrated WGAN inputs become unstable on the shrinking-bandwidth path: at $n = 10000$, half of the runs have multiple roots at one of the two bandwidths, one corrected estimate is negative, and corrected RMSE is 0.3064. These failures are retained rather than clipped or discarded. The calibration and both empirical comparators have no missing or multiple roots in this pilot. These results favor the small post-fit change over adding network or loss complexity, while retaining a clear empirical-only benchmark and the separate exact-target validity limitation.

7.9 Monotone generation and adaptive calibration

A second paired pilot asks whether a smaller structural generator and training-only weight selection can improve on fixed local calibration. The monotone generator represents the log-valuation quantile as a linear spline in $\Phi^{-1}(u)$, with eleven fixed knots, ten positive slopes, and one intercept. Linear tail extensions remove the finite support imposed by the original generator's tanh output. This is an eleven-parameter generator rather than a larger network. Monotonicity gives the exact distributional identity

$$\max_{1\le i\le N} Q_\gamma(U_i) \stackrel{d}{=} Q_\gamma(U^{1/N}), \qquad U, U_1, \ldots, U_N \sim \text{Uniform}(0,1), \tag{41}$$

where the bidder draws are independent. The right-hand side generates maxima directly. Local kernel moments use 32,768 deterministic midpoint quantiles of this distribution. These are numerical integration nodes, not additional observed auctions.

Each monotone generator starts from a data-only linear fit of log-maximum quantiles on $\Phi^{-1}(p^{1/N})$. This initialization favors lognormal structure; it is an eligible checkpoint, including step zero, rather than evidence of a benefit from adversarial training. The original critic, adversarial loss, gradient penalty, and 1,200 training steps are retained. The monotone generator learning rate is 5×10^{-4}; the original network keeps 2×10^{-5}. Architecture, tail extension, initialization, and deterministic integration change together, so this pilot does not identify their separate contributions. The fixed spline dimension and linear normal-quantile tails also remain approximation restrictions; unbounded support alone does not ensure tail accuracy.

For adaptive calibration, each outer training fold is split into an inner fitting sample (75 percent) and a validation sample (25 percent). Choose one shared CDF/density weight from $\{0, 0.25, 0.5, 1\}$ using standardized squared prediction errors of the observable kernel signals. Validation uses 21 prices between 0.6 and 1.4 times the inner fitting-sample direct-inversion reserve. Signal variances are estimated on that fitting sample and floored at 0.01. The generator is then refitted to the entire outer training fold. Only the untouched outer holdout enters its score. The same selected weight applies at all prices for a given bandwidth; different bandwidths may select different weights.

The four ablations vary generator (original or monotone) and mixing rule (fixed or adaptive), with identical observations and outer folds. The empirical-local DML and smoothed plug-in remain benchmarks. A lognormal maximum-order-statistic MLE supplies a correctly specified parametric benchmark in the lognormal design and a misspecified benchmark in the Weibull design. Its sandwich delta-method interval targets the fitted parametric reserve; under misspecification it need not cover the actual reserve. Adaptive mixing does not inherit the $O(n^{-1})$ fixed-weight argument: the local nuisance-rate and root assumptions require separate verification. All exact-reserve intervals below remain diagnostic.

The evaluation design fixes $N = 5$, $n \in \{500, 2000\}$, and ten new repetitions per cell. Valuations are either $\mathrm{LN}(0, 0.5^2)$ or Weibull with shape 1.5 and scale one. Development uses seed 20260929; final evaluation uses base seed 90260930 for lognormal and 20260930 for Weibull, which has a separate distribution offset. An initial lognormal cohort overlapped the prior calibration data because of additive seed indexing. It is archived separately and replaced in full, with no estimator or tuning changes, by the fresh lognormal cohort. No repetitions are removed based on their outcomes. The exact target uses the existing covariance-aware Richardson correction; the replication archive also reports the fixed-bandwidth target separately.

In the independent lognormal $n = 500$ cell, combined monotone/adaptive RMSE is 0.0525, compared with 0.0593 for original fixed calibration, 0.0591 for empirical DML, and 0.0530 for the smoothed plug-in. The paired MSE difference against the plug-in is -0.000052 with Monte Carlo standard error 0.000214, so the small ordering should not be interpreted as a stable advantage. The correctly specified lognormal MLE is considerably more accurate, with RMSE 0.0151, illustrating the information supplied by its parametric restriction.

At lognormal $n = 2000$, combined monotone/adaptive RMSE is 0.0318, against 0.0335 for original fixed calibration and empirical DML, and 0.0349 for the smoothed plug-in. The paired MSE difference against the plug-in is -0.000207 with Monte Carlo standard error

0.000236. Thus this cell also shows a favorable point estimate without resolving the ranking. Correctly specified MLE RMSE is 0.0073. Adaptive mixing with the original generator is worse than fixed calibration in all four cells; weight selection by itself is not the source of an established gain.

The Weibull cells caution against treating validation-selected mixing as an automatic improvement. At $n = 500$, the combined monotone/adaptive version has RMSE 0.0512 versus 0.0552 for original fixed calibration and 0.0503 for the smoothed plug-in. At $n = 2000$, its RMSE rises relative to these comparators: 0.0393 versus 0.0327 and 0.0328, respectively. Merely replacing the generator while retaining fixed calibration closely tracks the empirical estimator. The misspecified lognormal MLE has RMSE near 0.097 and zero observed actual-target coverage in both Weibull cells. This illustrates parametric misspecification, but cannot establish a WGAN-specific advantage because the empirical benchmarks also avoid it.

The fixed-calibration estimator therefore remains the main safeguarded procedure. The adaptive variants are reported as experiments, not as a validated replacement. Validation-based mixing and cross-fitting do not ensure dominance in reserve-price error, and ten repetitions leave substantial Monte Carlo uncertainty. The archive reports paired MSE differences with their Monte Carlo standard errors rather than selecting a method solely from the ordering of point estimates.

Table 8: Monotone generation and adaptive mixing: exact-reserve pilot

Design	Method	RMSE	Bias	Coverage	CI length
Lognormal, $n = 500$	Original fixed	0.0593	0.0270	90%	0.1855
	Monotone fixed	0.0575	0.0255	90%	0.1818
	Original adaptive	0.0872	0.0539	90%	0.2256
	Monotone adaptive	0.0525	0.0212	90%	0.1826
	Empirical DML	0.0591	0.0282	90%	0.1847
	Smooth plug-in	0.0530	0.0176	90%	0.1659
	Lognormal MLE	0.0151	-0.0064	100%	0.0495
Lognormal, $n = 2000$	Original fixed	0.0335	-0.0003	90%	0.1379
	Monotone fixed	0.0335	-0.0007	90%	0.1347
	Original adaptive	0.0558	0.0246	90%	0.1779
	Monotone adaptive	0.0318	-0.0010	90%	0.1306
	Empirical DML	0.0335	-0.0004	90%	0.1353
	Smooth plug-in	0.0349	-0.0061	80%	0.1286
	Lognormal MLE	0.0073	-0.0003	100%	0.0251
Weibull, $n = 500$	Original fixed	0.0552	-0.0080	90%	0.2393
	Monotone fixed	0.0534	-0.0092	100%	0.2295
	Original adaptive	0.0634	0.0331	100%	0.2589
	Monotone adaptive	0.0512	-0.0047	100%	0.2375
	Empirical DML	0.0529	-0.0088	90%	0.2239
	Smooth plug-in	0.0503	-0.0118	100%	0.2200
	Lognormal MLE	0.0966	-0.0958	0%	0.0529
Weibull, $n = 2000$	Original fixed	0.0327	0.0252	90%	0.1326
	Monotone fixed	0.0327	0.0251	90%	0.1312
	Original adaptive	0.0410	0.0329	90%	0.1409
	Monotone adaptive	0.0393	0.0315	90%	0.1370
	Empirical DML	0.0327	0.0251	90%	0.1304
	Smooth plug-in	0.0328	0.0247	90%	0.1290
	Lognormal MLE	0.0973	-0.0972	0%	0.0267

Notes: Ten paired repetitions per cell. All nonparametric methods use the same score, root rule, bandwidth path, and observations. Coverage and interval length concern nominal 95 percent intervals for the exact reserve. These are pilot diagnostics, not a coverage theorem. The MLE is correctly specified only in the lognormal design. Candidate validation risks, selected weights, and paired MSE Monte Carlo standard errors are included in the replication files.

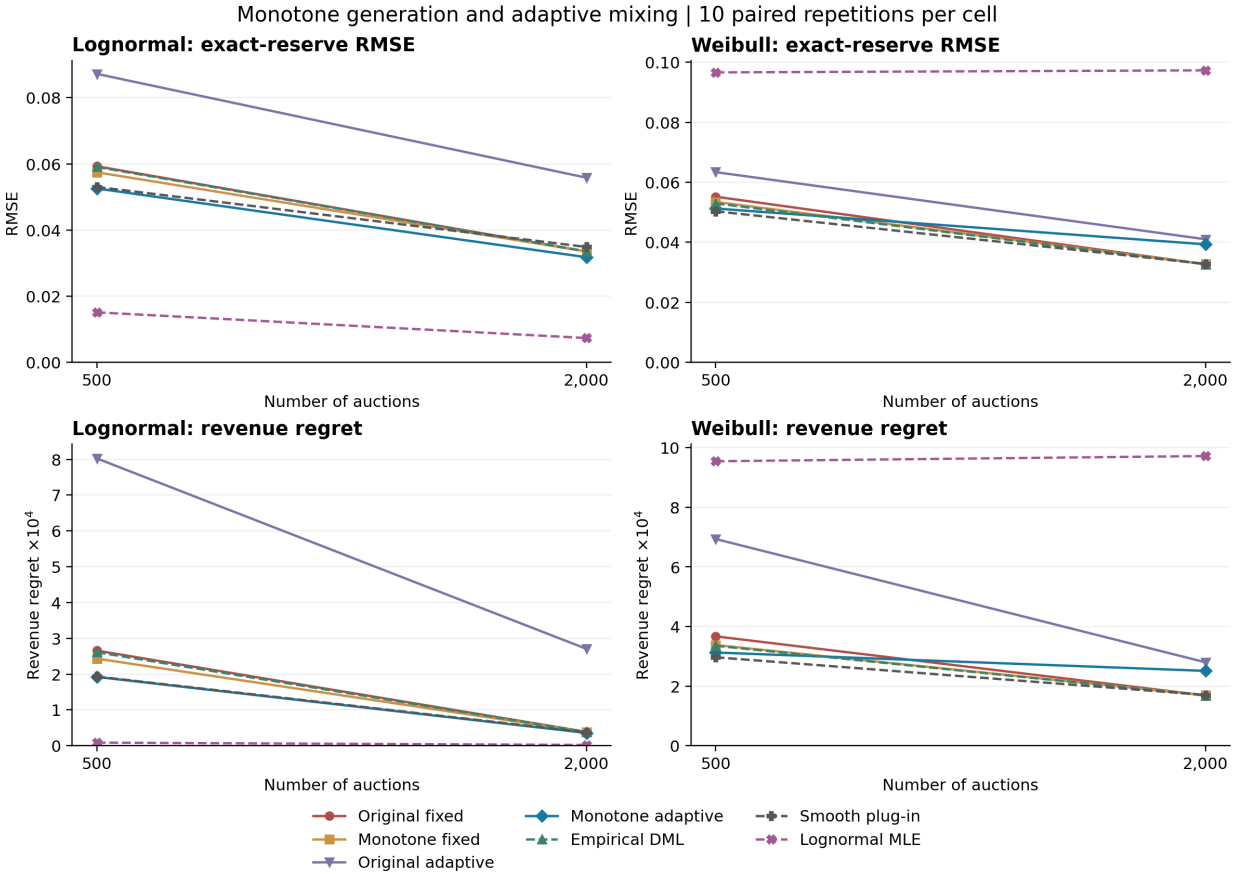


Figure 8: Exact-reserve error and revenue regret in the nested-validation pilot. Every method uses the same ten evaluation samples in each cell. No curve represents an oracle-selected tuning parameter.

7.9.1 Initialization versus adversarial training

The lognormal-shaped initialization could itself account for the preceding pilot's favorable point estimates. A retrospective paired ablation therefore reuses all forty evaluation datasets, without new tuning or sample selection. It holds the outer folds, inner split, numerical integration, score, and bandwidths fixed, and replaces adversarial training by the existing data-only initialization with zero optimizer steps. This linear normal-quantile initialization is a lognormal-family restriction, not an unrestricted learner. This is a mechanism diagnostic on already inspected samples, not a second independent confirmation.

Three comparisons distinguish training from weight selection. Fixed weights retain the fifty-pseudo-observation rule. Reselected weights allow initialized and trained generators to choose weights separately on the same inner validation data. Matched weights instead give the initialized outer nuisance the trained learner's weights. The last comparison isolates outer-nuisance training conditional on that weight-selection policy; it is not a cheap standalone estimator, since selecting its weights still requires training. All prior trained estimates are preserved. Thirty-two recomputed estimate and standard-error checks agree with the archive within 10^{-15}.

Table 9 shows that initialization alone nearly reproduces the lognormal adaptive results: RMSE is 0.0527 and 0.0314, compared with 0.0525 and 0.0318 after training. Under fixed calibration, all four training-induced RMSE changes are below 0.0004. These results do not identify adversarial learning as the source of the earlier lognormal gains. With separately selected weights, initialization also has lower point RMSE in both Weibull cells, although paired Monte Carlo uncertainty remains large.

The matched-weight comparison adds an important qualification. At Weibull $n = 2000$, training reduces RMSE from 0.0585 to 0.0393 when the trained selection rule's weights are held fixed. Its paired MSE difference is -0.00188, with Monte Carlo standard error 0.00108. Yet allowing the initialized learner to select its own weights gives RMSE 0.0366, while the smoothed plug-in remains more accurate at 0.0328. Averaged over all folds and distinct bandwidths in this cell, generated-component weights rise from 0.139 before training to 0.400 after training. Training and the induced weight response can therefore move the reserve error in different directions.

Training does improve observable-distribution fit under Weibull. Mean outer-holdout maximum-distribution W_1 falls from 0.0955 to 0.0780 at $n = 500$ and from 0.0657 to 0.0483 at $n = 2000$, using equal integration precision. These are descriptive means over thirty outer fits per cell; overlapping folds are not thirty independent replications. Holdout fit is used

only for reporting. Better global fit therefore does not establish improved reserve estimation under adaptive calibration. Figure 9 reports paired uncertainty: every displayed approximate Monte Carlo interval includes zero. The ten-repetition retrospective study motivates examining conservative mixing on fresh data, but does not validate such a modification or justify replacing the fixed-calibration safeguard.

Table 9: Exact-reserve RMSE before and after adversarial training

		Fixed weights		Reselected weights		Matched
Design	n	Init.	Trained	Init.	Trained	Init.
Lognormal	500	0.0577	0.0575	0.0527	0.0525	0.0521
Lognormal	2000	0.0335	0.0335	0.0314	0.0318	0.0316
Weibull	500	0.0538	0.0534	0.0486	0.0512	0.0485
Weibull	2000	0.0327	0.0327	0.0366	0.0393	0.0585

Notes: Ten paired repetitions per cell, reusing the preceding pilot's complete evaluation cohort. "Init." has zero adversarial updates. The matched initialized arm uses weights selected by the trained learner, so its comparator is the trained column under reselected weights. Fixed-bandwidth results, coverage, all raw estimates, validation risks, and cache provenance are included in the replication archive. No root failures occur in this ablation.

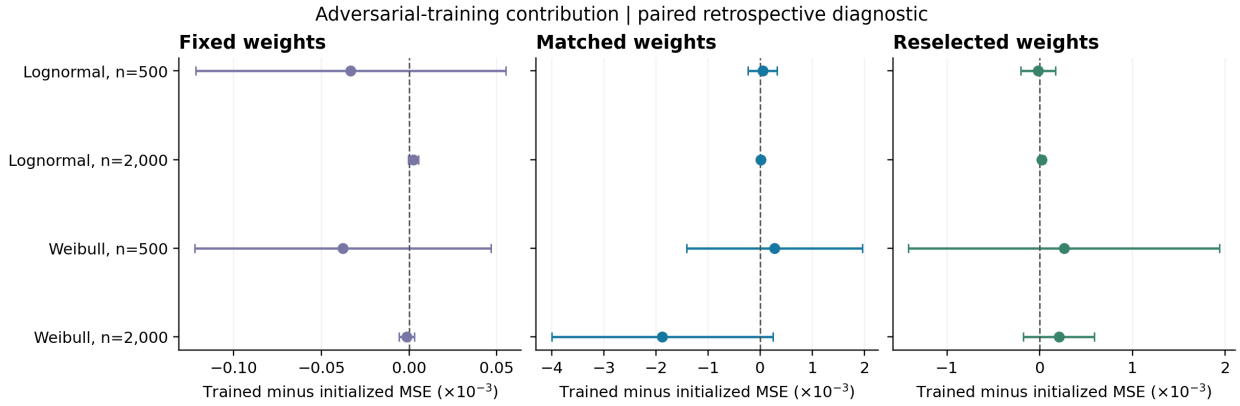


Figure 9: Paired adversarial-training effects on exact-reserve mean squared error. Negative values favor training. Bars are the mean paired squared-error difference plus or minus 1.96 Monte Carlo standard errors across ten replications, not confidence intervals for the reserve. Horizontal scales differ across panels. These retrospective, approximate uncertainty summaries are not a confirmatory dominance test.

7.10 Shrinking bandwidth and the exact reserve

The empirical-local ablation supports inference for a fixed regularized target. To examine shrinking-bandwidth behavior separately, I set

$$h_n = 0.15 \left(\frac{n}{500}\right)^{-\alpha}, \qquad \alpha \in \{0, 0.2, 0.35, 0.5\}, \tag{42}$$

and repeat the empirical-local estimator 200 times. The comparison holds the data seeds and score construction fixed. It reports coverage of both r_{h_n} and the exact reserve r_0, so decreasing regularization bias cannot be confused with improved score estimation.

Table 10: Shrinking-bandwidth paths for exact-reserve inference

n	Rule	h_n	$r_{h_n} - r_0$	RMSE to r_0	Mean SE	Coverage of r_0	Coverage of r_{h_n}	Multi-root
500	Fixed	0.1500	0.0600	0.0690	0.0335	0.6350	0.9000	0.0000
500	$n^{-0.2}$	0.1500	0.0600	0.0690	0.0335	0.6350	0.9000	0.0000
500	$n^{-0.35}$	0.1500	0.0600	0.0690	0.0335	0.6350	0.9000	0.0000
500	$n^{-0.5}$	0.1500	0.0600	0.0690	0.0335	0.6350	0.9000	0.0000
2000	Fixed	0.1500	0.0600	0.0648	0.0172	0.0100	0.9450	0.0000
2000	$n^{-0.2}$	0.1137	0.0409	0.0512	0.0256	0.5900	0.9500	0.0000
2000	$n^{-0.35}$	0.0923	0.0301	0.0482	0.0343	0.8200	0.9300	0.0000
2000	$n^{-0.5}$	0.0750	0.0217	0.0512	0.0451	0.8050	0.8850	0.0250
10000	Fixed	0.1500	0.0600	0.0621	0.0076	0.0000	0.9500	0.0000
10000	$n^{-0.2}$	0.0824	0.0252	0.0327	0.0172	0.6000	0.9300	0.0000
10000	$n^{-0.35}$	0.0526	0.0118	0.0354	0.0289	0.8200	0.8700	0.1050
10000	$n^{-0.5}$	0.0335	0.0052	0.0475	0.0298	0.7300	0.7300	0.6900

Notes: Two hundred common-seed repetitions per sample size. "Fixed" sets $h_n = 0.15$. Coverage columns refer to nominal 95 percent influence-function intervals. All paths share $h_{500} = 0.15$.

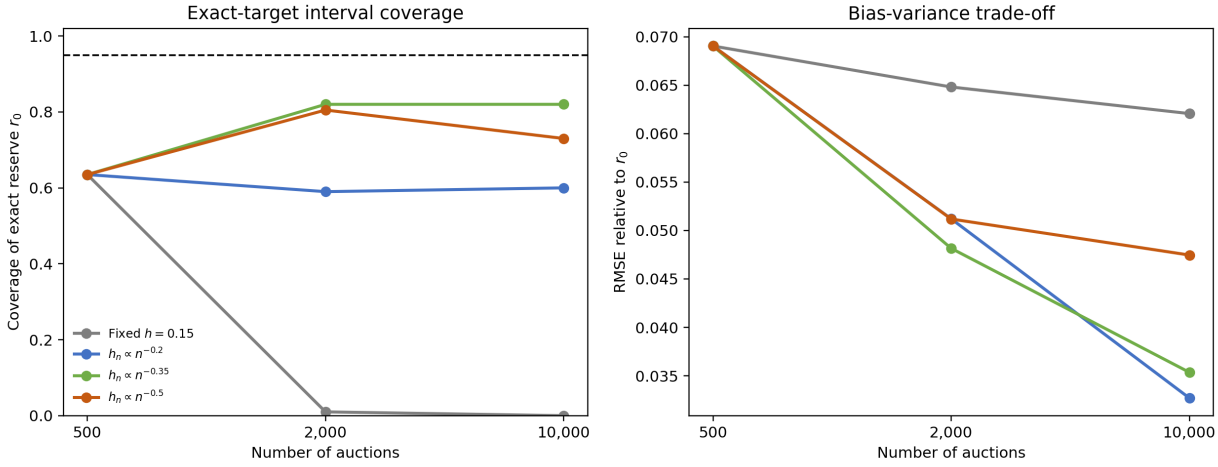


Figure 10: Exact-target coverage and RMSE along shrinking-bandwidth paths

Table 10 shows why fixed-h and exact-target inference must remain distinct. Fixed-h coverage of r_0 is 63.5 percent at $n = 500$, 1 percent at $n = 2000$, and zero at $n = 10000$, even though its coverage of r_h approaches 95 percent. The $n^{-0.2}$ rule reduces exact-target RMSE to 0.0327 at $n = 10000$, the smallest value among the four paths there, but exact coverage is only 60 percent.

The more aggressive $n^{-0.35}$ rule raises exact-target coverage to 82 percent at both $n = 2000$ and $n = 10000$, still below nominal. At $n = 10000$, 10.5 percent of samples have multiple score roots. Shrinking at $n^{-0.5}$ is worse: the multiple-root rate reaches 69 percent, exact-target RMSE rises to 0.0475, and coverage falls to 73 percent. Figure 10 summarizes the bias-variance and root-selection trade-off. No tested power rule simultaneously achieves low RMSE, nominal coverage of r_0, and stable roots in this sample range. Fixed-bandwidth inference for r_h is therefore the current defensible baseline; exact-reserve inference requires explicit bias correction or a confidence set that accounts for regularization and root uncertainty.

7.11 Covariance-aware bias correction

I next apply equation (33) to the two most relevant bandwidth paths, $\alpha \in \{0.2, 0.35\}$, using $\lambda = \sqrt{2}$. The two score estimates share the same observations and fold partition. Their observation-level influence values are combined before estimating the standard error, so the reported interval retains the covariance across bandwidths.

Table 11: Covariance-aware Richardson correction for the exact reserve

n	Estimator	h_n	Target $-r_0$	RMSE to r_0	Mean SE	Coverage of r_0	Target coverage	Multi-root
500	$n^{-0.2}$ DML	0.1500	0.0600	0.0690	0.0335	0.6350	0.9000	0.0000
500	$n^{-0.2}$ Richardson	0.1500	0.0269	0.0604	0.0500	0.8850	0.8950	0.0000
2000	$n^{-0.2}$ DML	0.1137	0.0409	0.0512	0.0256	0.5900	0.9500	0.0000
2000	$n^{-0.2}$ Richardson	0.1137	0.0162	0.0420	0.0386	0.9250	0.9400	0.0000
2000	$n^{-0.35}$ DML	0.0923	0.0301	0.0482	0.0343	0.8200	0.9300	0.0050
2000	$n^{-0.35}$ Richardson	0.0923	0.0104	0.0532	0.0514	0.8750	0.9050	0.0050
10000	$n^{-0.2}$ DML	0.0824	0.0252	0.0327	0.0172	0.6000	0.9300	0.0000
10000	$n^{-0.2}$ Richardson	0.0824	0.0080	0.0296	0.0255	0.8850	0.9300	0.0000
10000	$n^{-0.35}$ DML	0.0526	0.0118	0.0354	0.0289	0.8200	0.8700	0.3050
10000	$n^{-0.35}$ Richardson	0.0526	0.0023	0.0459	0.0413	0.8400	0.8500	0.3050

Notes: Two hundred common-seed repetitions. The DML row uses h_n; the Richardson row combines h_n and $\sqrt{2}h_n$. "Target" is the corresponding population regularized or extrapolated target. Multi-root records more than one grid crossing at either bandwidth. The two exponents coincide at $n = 500$, so that design is shown once.

For the stable $n^{-0.2}$ path, bias correction improves all three exact-target metrics. At

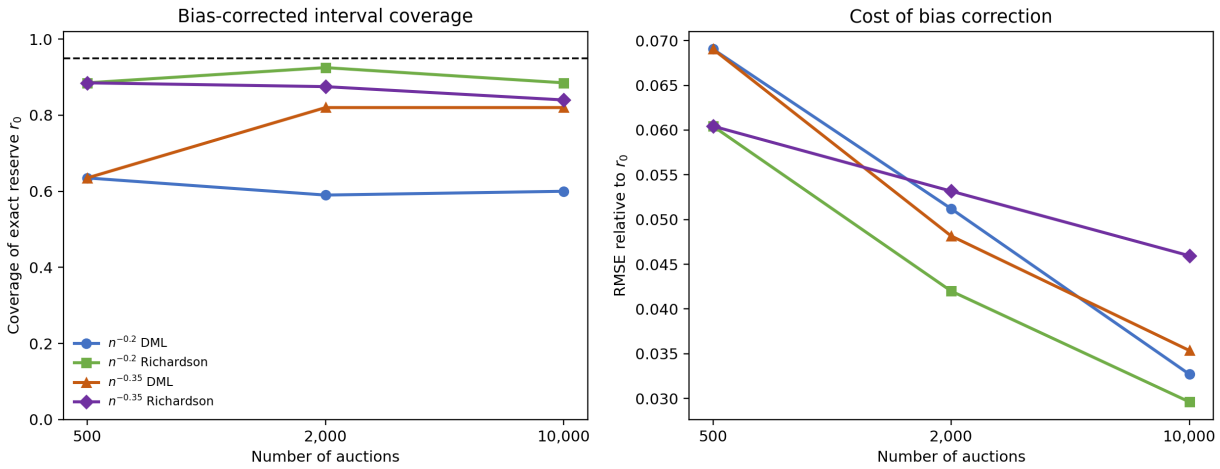


Figure 11: Exact-reserve performance after covariance-aware bias correction

$n = 2000$, coverage rises from 59.0 to 92.5 percent and RMSE falls from 0.0512 to 0.0420. At $n = 10000$, coverage rises from 60.0 to 88.5 percent and RMSE falls from 0.0327 to 0.0296. The extrapolated population-target bias there is 0.0080, compared with 0.0252 for the uncorrected target. No replication on this path has a missing or multiple root.

The faster $n^{-0.35}$ path does not benefit in the same way. Correction raises RMSE from 0.0482 to 0.0532 at $n = 2000$ and from 0.0354 to 0.0459 at $n = 10000$. At the largest sample size, 30.5 percent of replications have multiple roots at one of the two bandwidths, and corrected coverage is only 84.0 percent. Thus Richardson extrapolation is not a substitute for root stability. The $n^{-0.2}$ correction is the preferred exact-target prototype, but its 88.5 percent coverage at $n = 10000$ remains below nominal. This points to residual $O(h^4)$ bias, nonlinear root selection, or non-Gaussian score behavior as the next inferential problem.

7.12 Residual-bias envelope and root-robust set

I separate those explanations using the three-bandwidth construction in equations (39)–(40). The first row for each design is the covariance-aware Richardson Wald interval above. The second widens the selected-root interval by the estimated higher-order bias bound. The third takes the union over all empirical root-continuation chains. All three use the same 200 data seeds and $\lambda = \sqrt{2}$.

On the stable $n^{-0.2}$ path, the selected-root bias envelope attains 96.5 percent exact-reserve coverage at all three sample sizes. At $n = 10000$, this raises coverage from 88.5 percent for the Richardson Wald interval while increasing mean length from 0.0998 to 0.1279. Every

Table 12: Three-bandwidth residual-bias and root-robust confidence sets

n	α	Confidence set	Coverage of r_0	Target coverage	Mean length	Mean bias bound	Multi-root
500	0.2000	Richardson Wald	0.8850	0.8950	0.1959	0.0310	0.0000
500	0.2000	Selected-root bias envelope	0.9650	0.9800	0.2579	0.0310	0.0000
500	0.2000	Root-robust bias envelope	0.9650	0.9800	0.2579	0.0310	0.0000
2000	0.2000	Richardson Wald	0.9250	0.9400	0.1511	0.0214	0.0000
2000	0.2000	Selected-root bias envelope	0.9650	1.0000	0.1938	0.0214	0.0000
2000	0.2000	Root-robust bias envelope	0.9650	1.0000	0.1938	0.0214	0.0000
2000	0.3500	Richardson Wald	0.8750	0.9050	0.2013	0.0281	0.0050
2000	0.3500	Selected-root bias envelope	0.9750	0.9900	0.2575	0.0281	0.0050
2000	0.3500	Root-robust bias envelope	0.9750	0.9900	0.2612	0.0281	0.0050
10000	0.2000	Richardson Wald	0.8850	0.9300	0.0998	0.0140	0.0000
10000	0.2000	Selected-root bias envelope	0.9650	0.9850	0.1279	0.0140	0.0000
10000	0.2000	Root-robust bias envelope	0.9650	0.9850	0.1279	0.0140	0.0000
10000	0.3500	Richardson Wald	0.8400	0.8500	0.1620	0.0216	0.3050
10000	0.3500	Selected-root bias envelope	0.9150	0.9200	0.2051	0.0216	0.3050
10000	0.3500	Root-robust bias envelope	0.9150	0.9200	0.4064	0.0216	0.3050

Notes: "Selected-root bias envelope" adds equation (39) to both ends of the Richardson Wald interval. "Root-robust" is the union across all fine-bandwidth roots and their nearest-root continuation chains. Mean length is the total length of the union, excluding gaps. Multi-root records more than one grid crossing at any of the three bandwidths. The exponents coincide at $n = 500$, so that design is shown once.

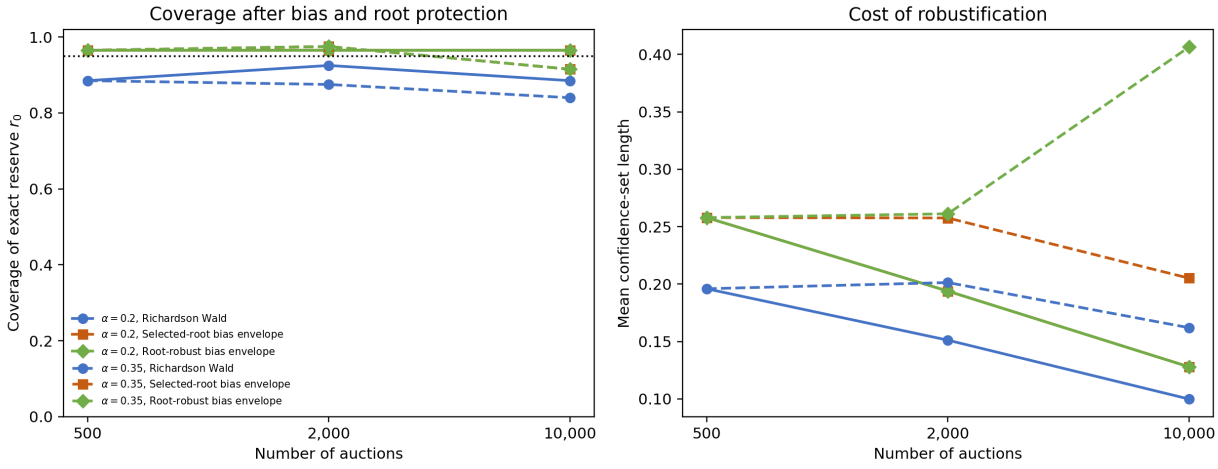


Figure 12: Coverage and length after residual-bias and root protection

replication has exactly one root, so the selected-root and root-robust sets coincide. This evidence assigns the stable-path undercoverage primarily to remaining higher-order bias or uncertainty in its estimate, rather than root selection.

The aggressive $n^{-0.35}$ path remains problematic. At $n = 10000$, 30.5 percent of samples have multiple roots. The bias envelope raises coverage from 84.0 to 91.5 percent, still below nominal. Taking the union over all root chains does not improve coverage further, but doubles mean set length from 0.2051 to 0.4064. Thus the omitted roots are mostly economically remote alternatives, not the source of the remaining coverage failures. The finite-sample evidence favors the $n^{-0.2}$ three-bandwidth envelope as the exact-target prototype, while retaining fixed-bandwidth inference as the formal baseline until the bias expansion and joint root process receive theoretical guarantees.

7.13 Competition and valuation distributions

The final experiment varies the number of bidders and the valuation family without adding another estimator or confidence-interval variant. It crosses $N \in \{3, 5, 10\}$ with the baseline lognormal distribution, a thicker-tailed lognormal distribution, and a two-component lognormal mixture. For each of the 18 design cells, Table 13 reports 200 replications of the empirical-local DML estimator with the same fixed Gaussian bandwidth $h = 0.15$ and three-fold cross-fitting. The formal inferential target is the cell-specific regularized root r_h. Exact-reserve RMSE is also reported for the DML estimate and direct order-statistic inversion so that smoothing bias is not hidden inside the inferential comparison.

For $N = 3$ or $N = 5$, target coverage ranges from 92.0 to 96.5 percent across the three distributions and two sample sizes. The regularized-target RMSE falls by roughly one half when the number of auctions rises from 2000 to 10000. These results show that the fixed-bandwidth theory is not specific to the baseline lognormal distribution. They also make clear that valid inference for r_h does not remove the economically separate approximation error $r_h - r_0$: that gap is largest in the mixture and increases with the number of bidders.

The difficult cells have $N = 10$. With 2000 auctions, the target coverage is 77.5, 78.0, and 78.5 percent in the baseline, heavy-tail, and mixture designs, respectively, and between 9.5 and 13.0 percent of samples have multiple roots. The expected number of observed auction maxima below the exact reserve is at most two in these cells and essentially zero in the baseline and mixture. With 10000 auctions, multiple roots disappear and coverage rises to 87.0, 93.5, and 89.5 percent. Thus greater competition makes a low reserve a more severe lower-tail extrapolation from maxima. Cross-fitting and orthogonality do not manufacture

Table 13: External validity of fixed-bandwidth local DML

DGP	N	n	$r_h - r_0$	DML RMSE to r_h	Coverage of r_h	DML RMSE to r_0	Direct RMSE to r_0	Multi-root
Baseline	3	2000	0.047	0.012	0.965	0.049	0.049	0.000
Baseline	3	10000	0.047	0.005	0.960	0.048	0.031	0.000
Baseline	5	2000	0.060	0.017	0.945	0.062	0.058	0.000
Baseline	5	10000	0.060	0.008	0.945	0.061	0.051	0.000
Baseline	10	2000	0.082	0.048	0.775	0.102	0.228	0.130
Baseline	10	10000	0.082	0.021	0.870	0.086	0.156	0.000
Heavy tail	3	2000	0.045	0.037	0.940	0.060	0.085	0.000
Heavy tail	3	10000	0.045	0.017	0.920	0.047	0.054	0.000
Heavy tail	5	2000	0.074	0.051	0.920	0.097	0.127	0.000
Heavy tail	5	10000	0.074	0.023	0.925	0.078	0.069	0.000
Heavy tail	10	2000	0.127	0.122	0.780	0.179	0.157	0.095
Heavy tail	10	10000	0.127	0.047	0.935	0.135	0.100	0.000
Mixture	3	2000	0.088	0.016	0.935	0.089	0.055	0.000
Mixture	3	10000	0.088	0.007	0.940	0.088	0.032	0.000
Mixture	5	2000	0.117	0.024	0.920	0.120	0.074	0.000
Mixture	5	10000	0.117	0.010	0.940	0.117	0.048	0.000
Mixture	10	2000	0.166	0.069	0.785	0.193	0.295	0.125
Mixture	10	10000	0.166	0.026	0.895	0.170	0.200	0.000

Notes: Each cell uses 200 replications, three folds, and $h = 0.15$. The heavy-tail design is lognormal with log standard deviation 0.8 rather than 0.5. The mixture places probability 0.7 on $\mathrm{LN}(-0.15, 0.35^2)$ and probability 0.3 on $\mathrm{LN}(0.55, 0.25^2)$. Coverage is for the regularized target r_h; exact-target columns use the global monopoly-price maximizer r_0. Multi-root is the fraction of samples with more than one score-grid crossing.

local information that the order statistic rarely contains. This is a transparent support limitation, not a generic failure of global distribution fit.

The exercise remains within symmetric independent private values and known N. It does not establish robustness to endogenous entry, auction heterogeneity, dependence across bidders, or structural misspecification.

8 Empirical Scope and Implementation

The current paper is a methodological Monte Carlo study rather than an application to a proprietary auction data set. An empirical implementation requires auction-level information that distinguishes the highest submitted bid from the transaction price, records the bidder count and auction format, and identifies whether a posted reserve censored participation or sale. Repeated auctions must be comparable after conditioning on observed item and market characteristics.

With those data, estimation proceeds within covariate cells or through conditional nuisance learners. Each training fold fits the structural distribution and the local score moments; held-out auctions enter only the orthogonal score. Bidder counts either index separate observation maps or enter the structural simulator. The fixed-bandwidth target should be reported first, followed by the bandwidth path, root count, and bias-aware exact-reserve interval. Counterfactual revenue calculations use the global structural fit, whereas uncertainty for the reserve uses the local score learner.

The maintained identification assumptions exclude endogenous entry, unobserved auction heterogeneity correlated with bidder values, affiliated private values, and confusion between the winning bid and transaction price. An application lacking information to address these issues would not validate the baseline model merely by matching the marginal distribution of observed prices.

9 Conclusion

This paper develops a semiparametric framework in which a structural WGAN-GP learns a latent valuation distribution while a cross-fitted orthogonal score targets a regularized reserve. Under fixed-bandwidth overlap, single-crossing and nuisance-rate conditions, the empirical-local and fixed-pseudo-count calibrated estimators are asymptotically linear and first-order equivalent to the same-kernel full-sample plug-in. These results do not establish

an efficiency gain from adversarial training. The score is locally orthogonal, not a two-sided doubly robust score, and standalone WGAN inference requires separate local-accuracy conditions.

The confidence-set construction preserves the nonlinear score and its candidate-dependent variance. It is distinct from the Wald interval and from the conservative interval envelope of its accepted values. The full set may be disconnected; retaining all components preserves the test-inversion definition. An envelope fills rejected gaps and cannot reduce coverage, but it can increase length and does not repair a lack of information. Reporting the original components, disconnection and boundary flags, and the envelope separately makes this distinction observable.

The independent validation provides limited rather than conclusive support for a small-sample coverage improvement. At $n = 500$, calibrated-GAN and empirical-local coverage rises from 91 to 93 percent under score inversion; at $n = 2000$, the constructions perform similarly near the nominal level. Two small-sample replications produce low-end accepted components associated with highly concentrated score variance. A sensitivity check that fixes the variance at the estimated root yields connected sets in the replay, but reduces small-sample coverage. The adopted response is therefore a minimal reporting safeguard, not another tuned estimator: retain the complete set, flag its shape and search-boundary contact, and provide an explicitly conservative envelope when a single interval is needed.

The archived experiments explain why local accuracy and information matter: good global distributional fit can coexist with reserve error, weakly observed tails, and inaccurate intervals. They motivate empirical local moments and calibration but do not demonstrate a stable reserve-estimation gain from the GAN. Their shrinking-bandwidth and residual-bias procedures remain diagnostic extensions, not substitutes for a theorem about the exact reserve. The strongest current formal claim is fixed-bandwidth inference under the stated assumptions. Further work must assess coverage across additional data-generating processes, justify any exact-reserve bandwidth rule, and establish a suitable empirical application without confusing price, bid and valuation data.

Data and Code Availability

The paper uses simulated data only; no proprietary or restricted dataset is analysed.

Archived replication materials are available at
`https://github.com/Shunyu1001/WGAN-DML-Auctions`.

Code, seeds, and numerical outputs for the updated confidence-set reporting and independent validation accompany this working-paper version.

Conflicts of Interest

The author declares that there are no conflicts of interest.

AI-Assisted Tools Declaration

OpenAI Codex was used to assist with drafting and editing text and code. The author reviewed the resulting text and code and accepts full responsibility for the content of the manuscript.

A Additional Derivations and Results

A.1 Proof of Proposition 1

Take expectations in equation (19). Since $\mathbb{E}_0 T_h = \eta_0$, the asserted identity follows. Differentiating that identity at $\eta = \eta_0$ cancels the derivative of m_h with the derivative of the residual term. Taylor expansion of $m_h(\theta, \eta_0)$ around η then gives

$$\mathbb{E}_0 \psi_h - m_h(\theta, \eta_0) = -\int_0^1 (1-t)(\eta_0 - \eta)^\mathsf{T} \nabla_\eta^2 m_h\{\theta, \eta + t(\eta_0 - \eta)\}(\eta_0 - \eta)\, dt. \tag{43}$$

On the stated region its Hessian is bounded, and its GG entry is zero because m_h is linear in G. This proves equation (22). If $A = A_h$, the Taylor expansion in G is exact and the remainder vanishes for every G. If only $G = G_h$, the generally nonzero AA remainder remains. Thus the result is not two-sided double robustness.

A.2 Proof of Proposition 2

For fixed h, the signals $T_h = (S_h, K_h)^\mathsf{T}$, their first θ derivatives, and their Lipschitz constants in θ are uniformly bounded by constants depending only on h. For example, $\partial_\theta S_h = K_h$ and $\partial_\theta K_h = -(\theta - B)K_h/h^2$. Applying Hoeffding's inequality to an n^{-1}-net of the compact interval Θ and bounding between grid points by the Lipschitz envelopes gives, for each

training fold,

$$\sup_{\theta\in\Theta} \|\widehat{\eta}_{E,h}^{(-k)}(\theta) - \eta_{0,h}(\theta)\| = O_p(\rho_n), \tag{44}$$

$$\sup_{\theta\in\Theta} \|\partial_\theta \widehat{\eta}_{E,h}^{(-k)}(\theta) - \partial_\theta \eta_{0,h}(\theta)\| = O_p(\rho_n). \tag{45}$$

The fold proportions and fixed K make the bounds uniform across folds. The same bounds hold for holdout averages of T_h and its first derivative. The generated averages and their first derivatives satisfy the same fixed envelopes, even if the generated distribution is misspecified or the number of draws is fixed. Since $w_k = O(n^{-1})$, replacing the empirical learner with its calibration changes both displays by only $O(n^{-1})$.

Uniform convergence and overlap imply $\inf_{k,\theta} \widehat{A}_h^{(-k)}(\theta) \geq a_h/2$ with probability approaching one, so clipping is inactive. On this event all functions of the nuisance in the score and its total derivative are uniformly bounded and continuous on a compact set. Substitution of the preceding uniform rates into the score and equation (28) proves uniform convergence of $\widehat{\Psi}_n$ and $\widehat{\Psi}'_n$.

For completeness, uniform stochastic control can also be expressed through the class $\{\psi_h(\cdot;\theta,a,g) : \theta \in \Theta, a \in [a_h/2, 1],\ g \in [0, (\sqrt{2\pi}h)^{-1}]\}$. This class has three real indices, bounded Lipschitz envelopes, and polynomial covering numbers. Its empirical process is therefore stochastically equicontinuous; see van der Vaart and Wellner [1996]. Estimated nuisance curves take values inside this class. This uniform argument, rather than a pointwise Chebyshev bound alone, justifies evaluations at nearby random roots.

A.3 Proof of Theorem 1

By Proposition 2, $\widehat{\Psi}_n \to_p m_{0,h}$ uniformly on Θ. Outside any fixed neighborhood of $\theta_{0,h}$, compactness and the unique-zero condition bound $|m_{0,h}|$ away from zero. Inside a sufficiently small neighborhood, uniform derivative convergence gives $\widehat{\Psi}'_n \leq -j_h$ with probability approaching one. Uniform convergence also preserves the positive and negative signs at the neighborhood endpoints and at the endpoints of Θ. The intermediate value theorem then gives exactly one interior downward root and no roots outside that neighborhood. Its bracket can be found on a grid containing the endpoints and refined. Clipping is inactive and the derivative rejection criterion is passed. Thus the failure probability tends to zero, and the selected approximate root is consistent. This derives root consistency rather than assuming it.

Write $P_{n,k}$ for the holdout empirical measure. At the fixed population root, decompose

$$\widehat{\Psi}_n(\theta_{0,h}) = \frac{1}{n}\sum_{i=1}^{n} \psi_h(B_i; \theta_{0,h}, \eta_{0,h}) + R_{1n} + R_{2n}, \tag{46}$$

where

$$R_{1n} = \sum_{k=1}^{K} \frac{|I_k|}{n}(P_{n,k} - P)\{\psi_h(\widehat{\eta}_h^{(-k)}) - \psi_h(\eta_{0,h})\}. \tag{47}$$

The common arguments $(B; \theta_{0,h})$ are suppressed in this display. Conditional on training observations and independent training randomization, the bracketed function is fixed for a holdout fold. Its conditional second moment is $O_p(\rho_n^2)$ by bounded nuisance derivatives. The fixed number of nondegenerate folds therefore gives $R_{1n} = O_p(\rho_n/\sqrt{n}) = o_p(n^{-1/2})$; no independence between different fold fits is required. Proposition 1 gives $R_{2n} = O_p(\rho_n^2) = o_p(n^{-1/2})$.

Let $e_n = \widehat{\Psi}_n(\widehat{\theta}_h)$, with $|e_n| \le \varepsilon_n = o(n^{-1/2})$. A mean-value expansion of the smooth sample score between the consistent root and $\theta_{0,h}$ gives

$$e_n = \widehat{\Psi}_n(\theta_{0,h}) + \widehat{\Psi}'_n(\theta_n^*)(\widehat{\theta}_h - \theta_{0,h}), \tag{48}$$

where uniform derivative convergence and continuity imply $\widehat{\Psi}'_n(\theta_n^*) \to_p J_h$. Substitution establishes equation (27). The oracle score is bounded for fixed h, has mean zero and positive variance, so the i.i.d. central limit theorem applies. Uniform nuisance convergence, root consistency, and bounded score envelopes also give consistency of the pooled second moment and $\widehat{J}_h$. Slutsky's theorem gives the variance and Wald statements. A centered difference with step tending to zero is consistent as well: it averages $\widehat{\Psi}'_n$ over a shrinking interval, whose difference from the continuous $m'_{0,h}$ tends uniformly to zero. This specializes the cross-fitted Z-estimation argument of Chernozhukov et al. [2018] without assuming root or Jacobian consistency.

A.4 Proof of Proposition 4

The fixed-target decomposition in the preceding proof gives

$$\sqrt{n}\,\widehat{\Psi}_n(\theta_{0,h}) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \psi_h(B_i; \theta_{0,h}, \eta_{0,h}) + o_p(1).$$

The summands have mean zero and variance $V_h > 0$. The fixed-h bounded envelope therefore gives the ordinary central limit theorem. Uniform nuisance convergence and the local CDF

lower bound also imply

$$\frac{1}{n}\sum_{i=1}^{n}\{\widehat{\psi}_{i,h}(\theta_{0,h}) - \psi_h(B_i;\theta_{0,h},\eta_{0,h})\}^2 = o_p(1).$$

Cauchy–Schwarz and the law of large numbers then give consistency of the pooled second moment. Since $\widehat{\Psi}_n(\theta_{0,h}) = O_p(n^{-1/2})$, centering the sample scores and using divisor $n-1$ does not change the limit: $\widehat{V}_n(\theta_{0,h}) \to_p V_h$. This argument does not treat the overlapping training folds as independent observations. Their estimated scores are compared with the common i.i.d. oracle scores, and the orthogonal remainder is controlled as in the preceding proof.

Slutsky's theorem yields $T_n(\theta_{0,h}) \Rightarrow \mathcal{N}(0,1)$. The probability that this statistic is undefined at the true parameter tends to zero by $V_h > 0$ and variance consistency; the point-estimation failure probability also tends to zero by Theorem 1. Consequently, treating failed point-estimation runs as noncoverage does not alter the limit. Membership of the true parameter in the continuous acceptance set is exactly its pointwise nonrejection event, which proves the first coverage statement. No correction for the number of candidate values is involved in that identity.

For every realized set, its interval envelope contains it. Hence

$$\Pr\{\theta_{0,h} \in \mathcal{E}_{n,1-\gamma}\} \geq \Pr\{\theta_{0,h} \in \mathcal{C}_{n,1-\gamma}\},$$

which proves the lower bound. Equality need not hold because the envelope includes gaps rejected by the score test. The proposition gives no claim of exact finite-sample coverage and no rate for errors from numerical detection of acceptance components. Those errors and the implementation's numerical failure rules are checked and reported separately. In particular, the pointwise argument alone does not bound every additional failure event from attempting to evaluate a complete finite-grid acceptance set.

A.5 Proof of Proposition 3

The full-sample kernel averages and their first derivatives have the same uniform rates. The preceding sign and derivative argument therefore supplies a unique consistent plug-in root with probability approaching one. At $\theta_{0,h}$, ordinary Taylor expansion in the two sample moments gives

$$m_h(\theta_{0,h},\widetilde{\eta}_h) = \nabla_\eta m_h(\theta_{0,h},\eta_{0,h})^{\mathsf{T}}(\widetilde{\eta}_h - \eta_{0,h}) + O_p(\rho_n^2) = \frac{1}{n}\sum_{i=1}^{n}\psi_h(B_i;\theta_{0,h},\eta_{0,h}) + o_p(n^{-1/2}). \tag{49}$$

The sample moment derivative converges uniformly to $m'_{0,h}$. Expansion at the plug-in root yields the same asymptotic linear representation as the cross-fitted empirical or calibrated root. Subtracting these representations proves the asserted first-order equivalence. The conclusion concerns the same fixed bandwidth and target; it does not compare an unsmoothed estimator with a smoothed one.

A.6 Proof of Proposition 5

The population Richardson combination satisfies

$$r_{\mathrm{R}}(h) = \frac{\lambda^2\theta_{0,h} - \theta_{0,\lambda h}}{\lambda^2 - 1} \tag{50}$$

$$= r_0 - \lambda^2 b_4 h^4 + o(h^4), \tag{51}$$

because the b_2h^2 terms cancel. Joint asymptotic linearity at the two bandwidths implies

$$\widehat{r}_{\mathrm{R}} - r_{\mathrm{R}}(h_n) = \frac{1}{n}\sum_{i=1}^{n} \frac{\lambda^2\varphi_{i,h_n} - \varphi_{i,\lambda h_n}}{\lambda^2 - 1} + o_p(s_{\mathrm{R},n}/\sqrt{n}). \tag{52}$$

The assumed triangular-array central limit theorem and covariance consistency studentize this sum. Condition (36) makes the remaining $O(h_n^4)$ population bias negligible relative to $s_{\mathrm{R},n}/\sqrt{n}$. If $s_{\mathrm{R},n} \asymp h_n^{-1/2}$, this condition is $\sqrt{nh_n}\,h_n^4 \to 0$, equivalently $nh_n^9 \to 0$; the additional condition $nh_n \to \infty$ supplies increasing local information.

A.7 Disconnected-set diagnostics and reporting

In the independent fixed-bandwidth validation, repetitions 29 and 50 at $n = 500$ produce disconnected score sets for both the calibrated and empirical-local estimators. Each contains a low-end component touching the prespecified lower search bound, 0.45. The original point-estimation rule still returns a unique accepted root. These four method–replication records are retained as successful computations with disconnection and lower-boundary flags; they are not recoded as failures.

The diagnosis reuses the saved data and generated samples without retraining. For both replications and all three estimators, grids with 281, 561, 1,121, and 2,241 points reproduce the detected set topology. Across these 24 comparisons, the largest endpoint change is below 2.4×10^{-12}. Direct centered variances and the implemented second-moment formula agree within 3.5×10^{-13}; 60-digit decimal aggregation of the same double-precision individual scores agrees within 2.3×10^{-13}. The latter checks summation accuracy, not arbitrary-precision

Table 14: Flagged score sets in the independent validation

Replication	Method	Full confidence set	Conservative envelope
29	Calibrated GAN	$[0.4500, 0.5836] \cup [0.7847, 0.9580]$	$[0.4500, 0.9580]$
29	Empirical DML	$[0.4500, 0.5654] \cup [0.7918, 0.9536]$	$[0.4500, 0.9536]$
50	Calibrated GAN	$[0.4500, 0.5376] \cup [0.6388, 0.9100]$	$[0.4500, 0.9100]$
50	Empirical DML	$[0.4500, 0.4891] \cup [0.6705, 0.9056]$	$[0.4500, 0.9056]$

Notes: Each row retains the complete detected acceptance set. The envelope is its smallest containing interval and includes the rejected gap. All four source sets are disconnected and touch the lower search boundary 0.45. These are reporting flags, not numerical failures. Endpoints are rounded outward to four decimal places for display; numerical membership uses the stored full-precision endpoints.

evaluation of the kernel functions. The affected CDF estimates remain above the clipping threshold. Thus these cases do not support a numerical-root or floating-point explanation for the disconnected sets.

At $\theta = 0.45$, the calibrated score means are approximately 1.798 and 1.364 in the two replications, whereas their score standard deviations are 24.557 and 16.604. The resulting studentized statistics, 1.637 and 1.836, fall below 1.96 even though the score means are far from zero. A single held-out observation accounts for 99.27 and 99.76 percent of the centered score sum of squares. In each case the smallest observed maximum is excluded from the training fold that supplies its nuisance estimate. A small training-fold CDF, amplified by the negative powers of A in the score, gives that held-out observation very high influence. The same pattern occurs with empirical-local moments, so it is not specific to the GAN.

This phenomenon concerns finite-sample information and the shape of the studentized score, not a second population optimum. When one score M dominates the sum and variance, $\overline{\psi} \approx M/n$ and $s_\psi \approx |M|/\sqrt{n}$, so the studentized statistic can be close to $\operatorname{sign}(M)$ rather than diverging at a distant candidate value. Oracle-moment comparisons reject the low-end candidate in both replications; they are attribution diagnostics and do not enter the reported estimator.

One simple sensitivity check fixes the score-variance denominator at the estimated root while retaining the nonlinear mean:

$$T_n^{\mathrm{fix}}(\theta) = \frac{\sqrt{n}\,\widehat{\Psi}_n(\theta)}{\sqrt{\widehat{V}_n(\widehat{\theta}_h)}}.$$

The same pointwise central limit and variance-consistency argument can give asymptotic coverage for its inversion, but does not ensure finite-sample improvement or global connect-

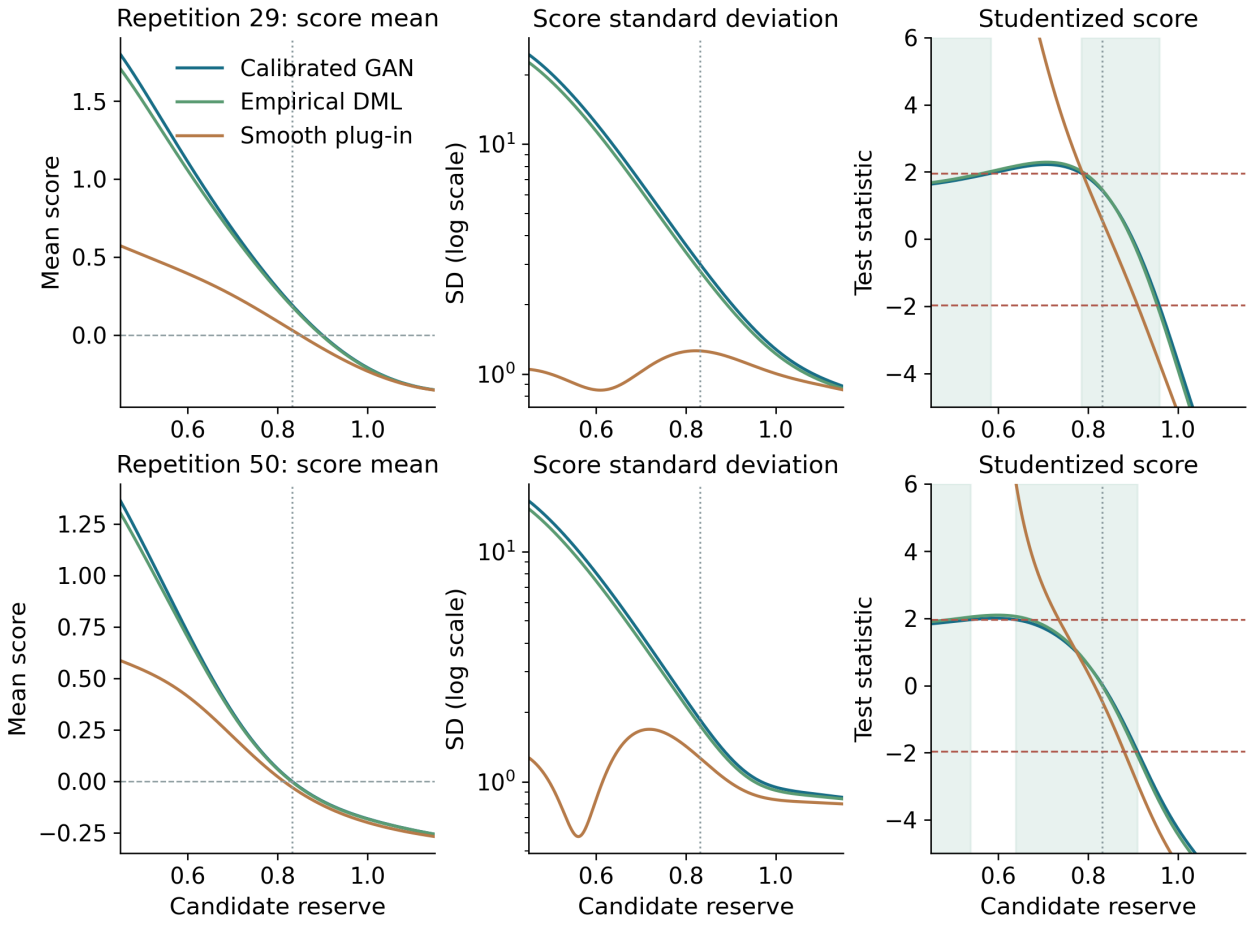


Figure 13: Score means, standard deviations, and studentized statistics in the two flagged replications. Shading marks the calibrated estimator's accepted components; horizontal dashed lines in the right panels mark ± 1.96 and vertical dotted lines mark $\theta_{0,h}$. The low-end component arises because the estimated score standard deviation increases, not because the mean score has an additional zero.

edness. On the saved validation replications this modification removes the disconnected components, yet lowers $n = 500$ coverage for both calibrated and empirical-local estimation from 93 to 90 percent. No new data are drawn for this sensitivity check, and it is not treated as another independent validation or adopted as the reporting rule.

Table 15: Fixed-denominator sensitivity on the saved validation samples

		Coverage (%)			Mean length	
n	Method	Wald	Full set	Fixed scale	Full set	Fixed scale
500	Calibrated GAN	91	93	90	0.14614	0.14494
500	Empirical DML	91	93	90	0.14208	0.14166
500	Smooth plug-in	91	92	92	0.12303	0.12512
2,000	Calibrated GAN	96	95	97	0.06741	0.06747
2,000	Empirical DML	96	95	96	0.06679	0.06684
2,000	Smooth plug-in	95	95	96	0.06426	0.06435

Notes: Each sample-size and method cell uses the same 100 replications as the independent validation. The fixed-denominator statistic uses the original estimated root's score standard deviation, without retraining or changing the point estimate. Coverage uses all 100 replications; length is the sum of accepted component lengths. This retrospective sensitivity comparison does not constitute a further independent confirmation. All three interval constructions have zero recorded computational failures in these samples.

The retained convention is therefore the full score set in equation (30), with explicit flags. Its optional envelope in equation (31) supplies a single interval without deleting accepted values. A disconnected set's reported total length is the sum of its component lengths, whereas the envelope includes rejected gaps. Any source-set disconnection and boundary flag remain visible when an envelope is shown. The lower bound is not increased after inspecting these replications, and no component or influential observation is removed to obtain a shorter interval.

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